Capgemini Exceller

Online Assessment - 10 Full Practice Sets

Sets 1-5 | Standard Analyst band - moderate difficulty

Sets H1-H5 | Senior Analyst band - medium to hard

Pseudo-code • Computer Fundamentals • English Communication
Game-Based Cognitive (Aon smartPredict) • Coding Round

300 pseudo-code and fundamentals questions • 90 English questions
50 game-based items • 20 coding problems • complete answer keys

How To Use This Book

This book holds ten complete practice sets modelled on the Capgemini Exceller online assessment. Sets 1 to 5 sit at the standard Analyst band. Sets H1 to H5 are deliberately harder and target the Senior Analyst band, the track that carries the higher package.

Each set is structured the same way as the live assessment.

Structure of every set

```
Section A   Pseudo-code (20 Q) + Computer Fundamentals (10 Q)     30 min
Section B   English Communication                                 10 min
Section C   Game-Based Cognitive (Aon smartPredict formats)        8-10 min
Section D   Coding Round (2 problems)                             45 min
```

Rules that mirror the live test

There is no negative marking anywhere in the assessment, so leave nothing blank. Every section is an elimination gate with its own cut-off, and you only reach the next section by clearing the current one. The live test is AI proctored on the Superset platform.

Section A cut-off: aim for 70 percent on Sets 1 to 5. On Sets H1 to H5, 55 to 65 percent is a strong result and is not a failing signal against real cut-offs.

Section B cut-off: aim for 65 percent.

Section C is scored on accuracy and response latency together, so time yourself per item, not only per section.

Section D determines the band. One problem fully passing maps to the middle band, both problems to the top band.

Reported band mapping

```
Both coding problems solved completely      7.50 LPA   (A5)
One coding problem solved completely        5.75 LPA   (A4)
All rounds cleared, partial coding          4.00 - 4.50 LPA   (A3)
```

Capgemini does not publish official cut-offs or band criteria. The figures above are aggregated candidate reports and should be treated as directional.

Languages for the coding round

C, C++ and Java. Reported guidance is that Python is not permitted in the Exceller coding round, though sources conflict on this point. Prepare in C, C++ or Java.

Notation Reference

Pseudo-code notation

Capgemini writes pseudo-code in a fixed notation. It is not a real language and no syntax knowledge is required. What is tested is your ability to trace variable state through the instructions.

```
Integer a, b, c            declaration
Integer Array[5]           array declaration
Set a = 5, b = 3           assignment
for(each i from 1 to n)    counting loop, inclusive of both bounds
for(each i from n down to 1)   descending loop
End for                    loop terminator
while(condition)            End while
repeat ... until(condition)     post-tested loop
if(cond)  else if(cond)  else   End if
MOD        integer remainder
/          INTEGER division - the fractional part is discarded
AND OR XOR bitwise operators
>>  <<     right and left bit shift
LENGTH(s)  string length
s[i]       character at index i, zero based
Print      output
End function name()         function terminator
```

Five pseudo-code patterns that dominate the bank

Accumulator loops - running sums and products over a bounded range. The trap is the exit condition; i less-than-or-equal-to n versus i less-than n changes the iteration count by one.

Modulo conditionals - nested if and else on divisibility or parity. The trap is losing track of which else pairs with which if.

String traversal - character iteration with LENGTH and indexing. The traps are treating LENGTH as the last valid index instead of length minus one, and confusing a character value with its position.

Number-theory routines - Euclidean GCD, digit sums, digit counts, base conversion. All built on repeated MOD and integer division, and they terminate in a handful of steps, so a full trace is always feasible.

Array swap and partial sort - you are given a starting array and asked for its state after exactly k iterations, not after the sort completes.

Game-based notation used in this book

The live games render coloured shapes and live animation. Printed pages cannot, so this book uses fixed ASCII tokens. The logic is identical.

```
STR = star        CIR = circle       TRI = triangle
SQR = square      PLS = plus         DIA = diamond

In grid questions:     X = cell holds a dot        . = empty cell
Arithmetic operators:  *  multiply    /  divide    +  add    -  subtract
```

switchChallenge - how the operator works

This is the deductive-logic game, and the one most worth drilling. Two rows of symbols are shown: an input row on top and an output row below. An operator is a string of digits. The digit in position k names which INPUT position supplies the symbol for OUTPUT position k.

```
Input     A  B  C  D
Operator  3  1  4  2
Output    C  A  D  B

Read it as:  output slot 1 takes input 3
             output slot 2 takes input 1
             output slot 3 takes input 4
             output slot 4 takes input 2
```

Levels escalate: one operator, then two operators with one of them hidden, then three with the middle hidden, then four. Six minutes, adaptive.

Two techniques carry the whole game. First-digit elimination: find the first output symbol in the input row, note its position, and discard every option that does not begin with that digit. Single-symbol tracking: follow one distinctive symbol forward from the input and backward from the output, and read off only the digit you need.

Learn to invert an operator directly. To invert 3142, ask where each input position lands: input 1 goes to slot 2, input 2 to slot 4, input 3 to slot 1, input 4 to slot 3, giving 2413. Several harder items in this book reduce to a single inversion.

gridChallenge

Alternating rounds. Memorise dot positions in a grid, then answer a spatial distractor - mirror symmetry, diagonal symmetry, or rotation - then recall the dots. The distractors exist specifically to overwrite your visual memory. Three to five memory rounds, adaptive, six minutes.

Convert positions to verbal labels such as row two far right rather than holding a picture. Verbal and visual memory use partly separate channels, so labelling survives the interference. Do not skip the spatial questions; failing them caps how far the adaptive algorithm will push you even with perfect recall.

digitChallenge

Complete an arithmetic equation by filling the blanks with digits 1 to 9, each digit used only once per equation. Standard operator precedence applies. Throughout this book at least one wrong option reaches the target value while reusing a digit, which is the real discriminator. Check distinctness before you check arithmetic.

motionChallenge

Live object tracking and complex control. Printed pages cannot animate, so this book uses a discrete-tick form: a ball moves one cell per tick on a grid and reflects off any wall it strikes. On a corner both components reverse at once. The underlying skill, holding a rule while updating state under time pressure, is the same.

Test Set 1

Standard Analyst band. Section A 30 min, Section B 10 min, Section C 8 min, Section D 45 min.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 8, b = 3
c = a MOD b
a = c << 2
Print a + b + c
```

A) 11 B) 13 C) 15 D) 8

Q2.

```
Integer p, q, r
Set p = 4, q = 2, r = 1
r = (p AND q) OR (r + 1)
Print r
```

A) 0 B) 1 C) 2 D) 3

Q3.

```
Integer sum, i
Set sum = 0
for(each i from 1 to 6)
  if(i MOD 2 == 0)
    sum = sum + i
  else
    sum = sum - i
  End if
End for
Print sum
```

A) 3 B) -3 C) 21 D) 9

Q4.

```
Integer solve(Integer n)
  Integer c
  Set c = 0
  while(n > 0)
    c = c + (n MOD 10)
    n = n / 10
  End while
  return c
End function solve()
Print solve(4728)
```

A) 17 B) 21 C) 11 D) 4

Q5.

```
Integer a, b
Set a = 20, b = 6
while(a != b)
  if(a > b)
    a = a - b
  else
    b = b - a
  End if
End while
Print a
```

A) 1 B) 2 C) 4 D) 6

Q6.

```
Integer Array[5]
Set Array = {3, 7, 2, 9, 4}
Integer i, m
Set m = Array[0]
for(each i from 1 to 4)
  if(Array[i] < m)
    m = Array[i]
  End if
End for
Print m
```

A) 2 B) 3 C) 9 D) 4

Q7.

```
Integer x, y
Set x = 5, y = 2
x = x / y
y = x * y
Print x, y
```

A) 2 4 B) 2.5 5 C) 2 5 D) 3 6

Q8.

```
String s
Set s = "PROGRAM"
Integer i, c
Set c = 0
for(each i from 0 to LENGTH(s) - 1)
  if(s[i] == 'R')
    c = c + 1
  End if
End for
Print c
```

A) 1 B) 2 C) 3 D) 0

Q9.

```
Integer n, f
Set n = 4, f = 1
while(n > 1)
  f = f * n
  n = n - 1
End while
Print f
```

A) 12 B) 24 C) 10 D) 6

Q10.

```
Integer a, b, c
Set a = 12, b = 5
c = a XOR b
Print c
```

A) 7 B) 9 C) 13 D) 4

Q11.

```
Integer i, j, c
Set c = 0
for(each i from 1 to 3)
  for(each j from i to 3)
    c = c + 1
  End for
End for
Print c
```

A) 6 B) 9 C) 3 D) 5

Q12.

```
Integer fun(Integer n)
  if(n <= 1)
    return n
  End if
  return fun(n - 1) + fun(n - 2)
End function fun()
Print fun(6)
```

A) 5 B) 8 C) 13 D) 6

Q13.

```
Integer n, r, rev
Set n = 132, rev = 0
while(n > 0)
  r = n MOD 10
  rev = rev * 10 + r
  n = n / 10
End while
Print rev
```

A) 231 B) 132 C) 213 D) 321

Q14.

```
Integer a, b, t
Set a = 3, b = 8
t = a
a = b
b = t
Print a - b
```

A) -5 B) 5 C) 11 D) 0

Q15.

```
Integer Array[6]
Set Array = {1, 2, 3, 4, 5, 6}
Integer i, s
Set s = 0
for(each i from 0 to 5)
  if(i MOD 2 == 0)
    s = s + Array[i]
  End if
End for
Print s
```

A) 9 B) 12 C) 21 D) 6

Q16.

```
Integer x, c
Set x = 1, c = 0
while(x < 100)
  x = x * 3
  c = c + 1
End while
Print c
```

A) 4 B) 5 C) 6 D) 3

Q17.

```
Integer a, b, c
Set a = 10, b = 20, c = 30
a = a + b
b = a - b
c = a - b
Print a, b, c
```

A) 30 10 20 B) 10 20 30 C) 30 20 10 D) 20 10 30

Q18.

```
Integer n, c, i
Set n = 29, c = 0
for(each i from 1 to n)
  if(n MOD i == 0)
    c = c + 1
  End if
End for
Print c
```

A) 1 B) 2 C) 29 D) 0

Q19.

```
Integer p, q
Set p = 16, q = 0
while(p > 1)
  p = p >> 1
  q = q + 1
End while
Print q
```

A) 3 B) 4 C) 5 D) 16

Q20.

```
Integer Array[5], i, t
Set Array = {5, 1, 4, 2, 8}
for(each i from 0 to 3)
  if(Array[i] > Array[i+1])
    t = Array[i]
    Array[i] = Array[i+1]
    Array[i+1] = t
  End if
End for
Print Array[4]
```

A) 5 B) 8 C) 2 D) 1

Section A - Computer Fundamentals (Q21 to Q30)

Q21. Time complexity of binary search on a sorted array of n elements?

A) O(n) B) O(log n) C) O(n log n) D) O(1)

Q22. Which OOP concept lets one function name work with different parameter lists?

A) Encapsulation B) Inheritance C) Polymorphism D) Abstraction

Q23. In C, the size of a pointer on a 64-bit system is:

A) 2 bytes B) 4 bytes C) 8 bytes D) Depends on the pointed-to type

Q24. Which SQL clause filters groups after aggregation?

A) WHERE B) HAVING C) GROUP BY D) ORDER BY

Q25. A stack follows which order?

A) FIFO B) LIFO C) Priority D) Random

Q26. Which normal form eliminates transitive functional dependency?

A) 1NF B) 2NF C) 3NF D) BCNF

Q27. Which is NOT one of the four necessary conditions for deadlock?

A) Mutual exclusion B) Hold and wait C) Preemption D) Circular wait

Q28. Which transport-layer protocol is connectionless?

A) TCP B) UDP C) IP D) HTTP

Q29. In a singly linked list of n nodes, deleting the last node costs:

A) O(1) B) O(log n) C) O(n) D) O(n squared)

Q30. Value of 7 AND 5 in C, using the bitwise operator?

A) 7 B) 5 C) 2 D) 12

Section B - English Communication (Q31 to Q40)

Q31. Neither the manager nor the engineers ____ available for the review.

A) is B) was C) were D) has been

Q32. Synonym of METICULOUS:

A) careless B) scrupulous C) hasty D) vague

Q33. Antonym of TRANSIENT:

A) fleeting B) brief C) permanent D) mobile

Q34. She is adept ____ handling client escalations.

A) in B) at C) on D) with

Q35. Find the error: "Each of the candidates have submitted their forms."

A) Each of B) have submitted C) their forms D) No error

Q36. Passive voice of "The team will complete the migration by Friday."

A) The migration will be completed by the team by Friday

B) The migration would be completed by the team by Friday

C) The migration is completed by the team by Friday

D) The migration has been completed by the team by Friday

Q37. "To cut corners" means:

A) Take a shortcut that lowers quality B) Negotiate hard

C) Finish early D) Divide work

Q38. Rearrange: (P) the deployment (Q) was postponed (R) due to an unresolved (S) security finding

A) PQRS B) QPRS C) PRQS D) SPQR

Q39. Indirect speech: He said, "I am reviewing the code."

A) He said that he is reviewing the code

B) He said that he was reviewing the code

C) He said that I was reviewing the code

D) He says that he was reviewing the code

Q40. Read the passage and answer.

```
Automated testing reduces regression defects, but teams that rely on it
exclusively often miss usability problems. Test suites verify what
developers anticipated; they cannot report what a confused user would
experience.
```

The passage implies automated testing is:

A) Unnecessary B) Sufficient on its own

C) Limited to anticipated scenarios D) More expensive than manual testing

Section C - Game-Based Cognitive (Q41 to Q44)

Q41. GeoSudo. Each row and each column contains CIR, TRI, SQR and PLS exactly once. What replaces the question mark?

```
CIR   TRI   SQR   PLS
TRI   CIR   PLS   SQR
SQR   PLS   CIR   TRI
PLS   SQR   TRI    ?
```

A) CIR B) TRI C) SQR D) PLS

Q42. digitChallenge. Insert operators so the equation holds.

```
5  _  3  _  2  =  13
```

A) + then * B) * then - C) - then * D) * then +

Q43. motionChallenge, discrete tick. A 4 by 4 grid. The ball starts at row 1, column 1 and the hole is at row 4, column 4. One orthogonal move per step. Cells (2,2) and (3,3) are blocked. Minimum number of steps?

A) 6 B) 7 C) 8 D) 3

Q44. Inductive reasoning. What is the next term?

```
2,  6,  12,  20,  30,  ?
```

A) 40 B) 42 C) 44 D) 36

Section D - Coding Round (45 minutes, C / C++ / Java)

Problem 1 - Second Largest Distinct Element

Given an integer array of size n, return the second largest distinct element. Return -1 if it does not exist. Single pass, O(n) time, O(1) extra space.

```
Input   [12, 35, 1, 10, 34, 1]        Output   34
Input   [7, 7, 7]                     Output   -1
```

Problem 2 - Pair Sum Count

Given an integer array and a target value, count the index pairs (i, j) with i less than j such that arr[i] + arr[j] equals the target. Handle duplicates correctly. O(n) time expected.

```
Input   arr = [1, 5, 7, -1, 5], target = 6      Output   3
        pairs: 1+5, 7+(-1), 1+5
```

Test Set 2

Standard Analyst band.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b
Set a = 9, b = 4
a = a MOD b
b = b MOD a
Print a + b
```

A) 0 B) 1 C) 5 D) 4

Q2.

```
Integer p, q, r
Set p = 6, q = 3, r = 2
r = (p OR q) AND (r << 1)
Print r
```

A) 0 B) 4 C) 7 D) 3

Q3.

```
Integer i, s
Set s = 1
for(each i from 1 to 4)
  s = s * 2
End for
Print s
```

A) 8 B) 16 C) 32 D) 24

Q4.

```
Integer fun(Integer n)
  if(n == 0)
    return 1
  End if
  return n * fun(n - 1)
End function fun()
Print fun(5) / fun(3)
```

A) 20 B) 24 C) 6 D) 40

Q5.

```
Integer n, c
Set n = 45, c = 0
while(n > 0)
  if((n MOD 10) MOD 2 == 1)
    c = c + 1
  End if
  n = n / 10
End while
Print c
```

A) 0 B) 1 C) 2 D) 45

Q6.

```
Integer Array[5], i, s
Set Array = {2, 4, 6, 8, 10}
Set s = 0
for(each i from 0 to 4)
  s = s + Array[i] * i
End for
Print s
```

A) 60 B) 80 C) 30 D) 100

Q7.

```
String s
Set s = "LEVEL"
Integer i, f
Set f = 1
for(each i from 0 to LENGTH(s)/2 - 1)
  if(s[i] != s[LENGTH(s) - 1 - i])
    f = 0
  End if
End for
Print f
```

A) 0 B) 1 C) 5 D) 2

Q8.

```
Integer x, y, z
Set x = 2, y = 3, z = 4
x = y
y = z
z = x
Print x, y, z
```

A) 3 4 3 B) 2 3 4 C) 4 3 2 D) 3 4 2

Q9.

```
Integer a, b, i
Set a = 1, b = 1
for(each i from 1 to 5)
  a = a + b
  b = a - b
End for
Print a, b
```

A) 13 8 B) 8 5 C) 21 13 D) 12 7

Q10.

```
Integer n, c
Set n = 100, c = 0
while(n > 0)
  n = n / 2
  c = c + 1
End while
Print c
```

A) 6 B) 7 C) 8 D) 5

Q11.

```
Integer i, j, s
Set s = 0
for(each i from 1 to 3)
  for(each j from 1 to 3)
    if(i == j)
      s = s + i * j
    End if
  End for
End for
Print s
```

A) 14 B) 6 C) 36 D) 9

Q12.

```
Integer Array[4], i, j, t
Set Array = {7, 3, 9, 1}
for(each i from 0 to 2)
  for(each j from 0 to 2 - i)
    if(Array[j] > Array[j+1])
      t = Array[j]
      Array[j] = Array[j+1]
      Array[j+1] = t
    End if
  End for
End for
Print Array[0], Array[3]
```

A) 1 9 B) 7 1 C) 9 1 D) 3 9

Q13.

```
Integer a
Set a = 5
if(a > 3)
  if(a > 7)
    Print "HIGH"
  else
    Print "MID"
  End if
else
  Print "LOW"
End if
```

A) HIGH B) MID C) LOW D) No output

Q14.

```
Integer n, s
Set n = 6, s = 0
while(n != 0)
  s = s + n
  n = n - 2
End while
Print s
```

A) 12 B) 21 C) 6 D) Infinite loop

Q15.

```
Integer x, y
Set x = 7, y = 2
Print x / y, x MOD y
```

A) 3 1 B) 3.5 1 C) 3 0 D) 2 3

Q16.

```
String s, r
Integer i
Set s = "DATA", r = ""
for(each i from LENGTH(s) - 1 down to 0)
  r = r + s[i]
End for
Print r
```

A) DATA B) ATAD C) AATD D) TADA

Q17.

```
Integer p, q
Set p = 3, q = 5
p = p XOR q
q = p XOR q
p = p XOR q
Print p, q
```

A) 5 3 B) 3 5 C) 6 3 D) 0 0

Q18.

```
Integer n, c
Set n = 8, c = 0
while(n > 0)
  if(n MOD 2 == 1)
    c = c + 1
  End if
  n = n / 2
End while
Print c
```

A) 1 B) 3 C) 4 D) 0

Q19.

```
Integer Array[5], i, j, t
Set Array = {1, 2, 3, 4, 5}
Set i = 0, j = 4
while(i < j)
  t = Array[i]
  Array[i] = Array[j]
  Array[j] = t
  i = i + 1
  j = j - 1
End while
Print Array[1]
```

A) 2 B) 4 C) 3 D) 1

Q20.

```
Integer a, b, c
Set a = 15, b = 4, c = 0
while(a >= b)
  a = a - b
  c = c + 1
End while
Print c, a
```

A) 3 3 B) 4 0 C) 3 4 D) 2 7

Section A - Computer Fundamentals (Q21 to Q30)

Q21. Worst-case time complexity of QuickSort?

A) O(n log n) B) O(n squared) C) O(log n) D) O(n)

Q22. Which is achieved through virtual functions in C++?

A) Compile-time polymorphism B) Runtime polymorphism

C) Encapsulation D) Overloading

Q23. In Java, which keyword prevents a class from being subclassed?

A) static B) private C) final D) abstract

Q24. Which join returns all left-table rows plus matched right-table rows?

A) INNER JOIN B) LEFT OUTER JOIN C) RIGHT OUTER JOIN D) CROSS JOIN

Q25. A queue built from two stacks: amortised dequeue cost is:

A) O(1) B) O(n) C) O(log n) D) O(n squared)

Q26. Which scheduling algorithm can cause starvation?

A) FCFS B) Round Robin C) Priority scheduling D) None

Q27. Default port for HTTPS?

A) 21 B) 80 C) 443 D) 8080

Q28. Which structure drives BFS traversal?

A) Stack B) Queue C) Heap D) Linked list

Q29. A dangling pointer in C refers to:

A) A NULL address B) Memory already freed C) An array D) A function

Q30. In ACID, the letter I stands for:

A) Integrity B) Isolation C) Indexing D) Immutability

Section B - English Communication (Q31 to Q40)

Q31. The report, along with its appendices, ____ submitted yesterday.

A) were B) was C) have D) are

Q32. Synonym of PRAGMATIC:

A) idealistic B) practical C) theoretical D) reckless

Q33. Antonym of AUGMENT:

A) expand B) diminish C) supplement D) enhance

Q34. The migration is contingent ____ client approval.

A) to B) on C) for D) with

Q35. Find the error: "He suggested to postpone the release."

A) He suggested B) to postpone C) the release D) No error

Q36. Active voice of "The bug was reported by the QA team."

A) The QA team reports the bug B) The QA team reported the bug

C) The QA team has reported the bug D) The bug reports the QA team

Q37. "Back to the drawing board" means:

A) Start planning again after failure B) Draw diagrams

C) Revisit old files D) Delegate

Q38. Rearrange: (P) before the sprint (Q) the backlog (R) must be groomed (S) begins

A) QRPS B) PQRS C) QRSP D) PSQR

Q39. Indirect speech: She asked, "Where is the config file?"

A) She asked where the config file is

B) She asked where the config file was

C) She asked where was the config file

D) She asks where the config file was

Q40. Read the passage and answer.

```
Code review catches defects, but its greater value is knowledge transfer.
A reviewer who reads unfamiliar modules acquires context that becomes
valuable when the original author leaves the team.
```

The primary benefit emphasised is:

A) Defect detection B) Faster releases

C) Distribution of knowledge D) Reduced documentation

Section C - Game-Based Cognitive (Q41 to Q44)

Q41. GeoSudo. What replaces the question mark?

```
TRI   SQR   PLS   CIR
SQR   PLS   CIR   TRI
PLS   CIR   TRI   SQR
CIR   TRI   SQR    ?
```

A) CIR B) TRI C) SQR D) PLS

Q42. digitChallenge. Insert operators so the equation holds.

```
8  _  4  _  3  =  5
```

A) / then + B) * then - C) - then + D) + then /

Q43. Inductive reasoning. What is the next term?

```
3,  8,  15,  24,  35,  ?
```

A) 44 B) 46 C) 48 D) 50

Q44. switchChallenge style rule application. Switch A doubles the value. Switch B subtracts 3. Switch C halves it using integer division. Starting at 10, apply A, B, C, B in that order. Final value?

A) 4 B) 5 C) 7 D) 8

Section D - Coding Round (45 minutes)

Problem 1 - Move Zeros

Given an integer array, shift every 0 to the end while preserving the relative order of the non-zero elements. Modify the array in place. O(n) time, O(1) extra space.

```
Input   [0, 1, 0, 3, 12]        Output   [1, 3, 12, 0, 0]
```

Problem 2 - Longest Substring Without Repeating Characters

Given a string, return the length of the longest substring that contains no repeated character. Sliding window, O(n).

```
Input   "abcabcbb"     Output   3
Input   "bbbbb"        Output   1
Input   "pwwkew"       Output   3
```

Test Set 3

Standard Analyst band.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 11, b = 4
c = a / b + a MOD b
Print c
```

A) 5 B) 2 C) 3 D) 7

Q2.

```
Integer p, q
Set p = 5, q = 3
p = p << q
Print p
```

A) 15 B) 40 C) 8 D) 20

Q3.

```
Integer i, s
Set s = 0
for(each i from 10 down to 1)
  if(i MOD 3 == 0)
    s = s + i
  End if
End for
Print s
```

A) 18 B) 12 C) 21 D) 9

Q4.

```
Integer fun(Integer a, Integer b)
  if(b == 0)
    return 1
  End if
  return a * fun(a, b - 1)
End function fun()
Print fun(3, 4)
```

A) 12 B) 27 C) 81 D) 64

Q5.

```
Integer n, c
Set n = 1024, c = 0
while(n > 1)
  n = n / 4
  c = c + 1
End while
Print c
```

A) 4 B) 5 C) 6 D) 10

Q6.

```
Integer Array[6], lo, hi, mid, key, pos
Set Array = {4, 8, 15, 16, 23, 42}
Set lo = 0, hi = 5, key = 23, pos = -1
while(lo <= hi)
  mid = (lo + hi) / 2
  if(Array[mid] == key)
    pos = mid
    break
  else if(Array[mid] < key)
    lo = mid + 1
  else
    hi = mid - 1
  End if
End while
Print pos
```

A) 3 B) 4 C) 5 D) -1

Q7.

```
String s
Integer i, c
Set s = "ENGINEER", c = 0
for(each i from 0 to LENGTH(s) - 1)
  if(s[i] == 'E')
    c = c + 1
  End if
End for
Print c
```

A) 2 B) 3 C) 4 D) 8

Q8.

```
Integer a, b
Set a = 6, b = 2
while(b <= a)
  b = b * 2
End while
Print b
```

A) 4 B) 8 C) 6 D) 16

Q9.

```
Integer x, y, z
Set x = 4, y = 5
if(x > y)
  z = x
else
  z = y
End if
Print z * 2
```

A) 8 B) 10 C) 9 D) 20

Q10.

```
Integer n, s, t
Set n = 3, s = 0
for(each t from 1 to n)
  s = s + t * t * t
End for
Print s
```

A) 14 B) 36 C) 27 D) 6

Q11.

```
Integer Array[5], i, s1, s2
Set Array = {10, 20, 30, 40, 50}
Set s1 = 0, s2 = 0
for(each i from 0 to 4)
  if(i < 2)
    s1 = s1 + Array[i]
  else
    s2 = s2 + Array[i]
  End if
End for
Print s2 - s1
```

A) 60 B) 90 C) 120 D) 30

Q12.

```
Integer a, b, c
Set a = 2, b = 3, c = 4
a = a * b
b = b + c
c = a - b
Print a, b, c
```

A) 6 7 -1 B) 6 7 1 C) 2 3 4 D) 5 7 -2

Q13.

```
Integer i, c
Set c = 0
for(each i from 1 to 20)
  if(i MOD 4 == 0 AND i MOD 6 == 0)
    c = c + 1
  End if
End for
Print c
```

A) 0 B) 1 C) 2 D) 3

Q14.

```
Integer n, d, c
Set n = 1010, c = 0, d = 1
while(n > 0)
  c = c + (n MOD 10) * d
  d = d * 2
  n = n / 10
End while
Print c
```

A) 10 B) 1010 C) 5 D) 2

Q15.

```
Integer p, q, r
Set p = 9, q = 6
r = p AND q
Print r
```

A) 0 B) 15 C) 9 D) 6

Q16.

```
String s
Integer i, c
Set s = "aAbB", c = 0
for(each i from 0 to LENGTH(s) - 1)
  if(s[i] >= 'a' AND s[i] <= 'z')
    c = c + 1
  End if
End for
Print c
```

A) 1 B) 2 C) 3 D) 4

Q17.

```
Integer i, j
Set j = 0
for(each i from 1 to 5)
  if(i == 3)
    continue
  End if
  j = j + i
End for
Print j
```

A) 15 B) 12 C) 10 D) 6

Q18.

```
Integer Array[5], i, c
Set Array = {3, 3, 5, 5, 7}
Set c = 0
for(each i from 0 to 3)
  if(Array[i] == Array[i+1])
    c = c + 1
  End if
End for
Print c
```

A) 1 B) 2 C) 3 D) 0

Q19.

```
Integer a, b, t, i
Set a = 0, b = 1
for(each i from 1 to 6)
  t = a + b
  a = b
  b = t
End for
Print a
```

A) 5 B) 8 C) 13 D) 21

Q20.

```
Integer n, m, c
Set n = 30, m = 12, c = 1
while(m != 0)
  c = n MOD m
  n = m
  m = c
End while
Print n * 2
```

A) 6 B) 12 C) 24 D) 4

Section A - Computer Fundamentals (Q21 to Q30)

Q21. Auxiliary space complexity of merge sort on an array of n elements?

A) O(1) B) O(log n) C) O(n) D) O(n log n)

Q22. Which principle hides internal state behind an interface?

A) Inheritance B) Encapsulation C) Polymorphism D) Composition

Q23. In C++, which is NOT inherited by a derived class?

A) public members B) protected members C) constructors D) member functions

Q24. Which statement removes all rows, keeps the table structure, and in most DBMSs cannot be rolled back?

A) DELETE B) DROP C) TRUNCATE D) ALTER

Q25. Maximum nodes in a binary tree of height h, with the root at height 0?

A) 2 to the power h B) 2 to the power (h+1), minus 1

C) 2h + 1 D) h squared

Q26. Thrashing in an operating system is caused by:

A) Excessive page faults B) CPU overheating C) Deadlock D) Disk fragmentation

Q27. Which OSI layer handles routing?

A) Data link B) Network C) Transport D) Session

Q28. Which BST traversal yields sorted order?

A) Preorder B) Inorder C) Postorder D) Level order

Q29. In Java, the mutable and non-thread-safe alternative to String is:

A) StringBuffer B) StringBuilder C) CharArray D) StringJoiner

Q30. A candidate key not chosen as the primary key is called:

A) Foreign key B) Alternate key C) Super key D) Composite key

Section B - English Communication (Q31 to Q40)

Q31. Not only the developers but also the architect ____ present at the demo.

A) were B) are C) was D) have been

Q32. Synonym of SUCCINCT:

A) verbose B) concise C) elaborate D) hesitant

Q33. Antonym of OBSOLETE:

A) outdated B) archaic C) current D) rare

Q34. They complied ____ the new security policy.

A) to B) with C) on D) for

Q35. Find the error: "The number of open tickets have increased sharply."

A) The number of B) have increased C) sharply D) No error

Q36. Passive of "Someone has deleted the log files."

A) The log files have been deleted B) The log files were deleted

C) The log files are deleted D) The log files had been deleted

Q37. "In the loop" means:

A) Stuck repeating B) Kept informed C) Writing loops D) Out of scope

Q38. Rearrange: (P) unless the root cause (Q) the incident (R) is identified (S) will recur

A) PRQS B) QSPR C) PQRS D) QRPS

Q39. Indirect speech: The lead said, "We deployed the patch last night."

A) The lead said that they deployed the patch last night

B) The lead said that they had deployed the patch the previous night

C) The lead said that we had deployed the patch last night

D) The lead says that they deployed the patch last night

Q40. Read the passage and answer.

```
Microservices reduce coupling between teams but increase operational
surface area. An organisation without mature monitoring often trades a
manageable monolith for a distributed system it cannot observe.
```

The author's caution is that microservices:

A) Always perform worse B) Require operational maturity to pay off

C) Eliminate coupling entirely D) Are cheaper to run

Section C - Game-Based Cognitive (Q41 to Q44)

Q41. GeoSudo. What replaces the question mark?

```
PLS   CIR   TRI   SQR
CIR   TRI   SQR   PLS
TRI   SQR   PLS   CIR
SQR   PLS   CIR    ?
```

A) CIR B) TRI C) SQR D) PLS

Q42. digitChallenge. Insert operators so the equation holds.

```
6  _  2  _  7  =  5
```

A) / then + B) * then - C) + then - D) - then *

Q43. Inductive reasoning. What is the next term?

```
5,  11,  23,  47,  ?
```

A) 89 B) 92 C) 94 D) 95

Q44. Rule application. Switch X adds 5. Switch Y multiplies by 3. Starting at 4, apply Y, X, Y in that order. Final value?

A) 27 B) 39 C) 51 D) 57

Section D - Coding Round (45 minutes)

Problem 1 - Remove Duplicates from Sorted Array

Given a sorted array, remove duplicates in place so each element appears once, and return the new length. Elements beyond the returned length do not matter. O(n) time, O(1) space.

```
Input   [1, 1, 2, 2, 2, 3, 4, 4]
Output  4, and the array begins [1, 2, 3, 4]
```

Problem 2 - Maximum Subarray Sum

Given an integer array that may contain negatives, return the largest sum obtainable from any contiguous non-empty subarray. O(n) single pass.

```
Input   [-2, 1, -3, 4, -1, 2, 1, -5, 4]     Output   6
Input   [-3, -1, -7]                        Output   -1
```

Test Set 4

Standard Analyst band.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b
Set a = 13, b = 5
Print a MOD b, a / b
```

A) 3 2 B) 2 3 C) 3 2.6 D) 2 3.0

Q2.

```
Integer p
Set p = 40
p = p >> 2
p = p << 1
Print p
```

A) 40 B) 20 C) 10 D) 80

Q3.

```
Integer i, s
Set s = 0
for(each i from 1 to 5)
  s = s + i * (i + 1)
End for
Print s
```

A) 55 B) 70 C) 30 D) 45

Q4.

```
Integer fun(Integer n)
  if(n < 2)
    return n
  End if
  return fun(n / 2) + (n MOD 2)
End function fun()
Print fun(13)
```

A) 2 B) 3 C) 4 D) 13

Q5.

```
Integer n, c
Set n = 7, c = 0
while(n != 1)
  if(n MOD 2 == 0)
    n = n / 2
  else
    n = 3 * n + 1
  End if
  c = c + 1
End while
Print c
```

A) 11 B) 16 C) 17 D) 7

Q6.

```
Integer Array[6], i, mx, mn
Set Array = {5, 2, 9, 1, 7, 3}
Set mx = Array[0], mn = Array[0]
for(each i from 1 to 5)
  if(Array[i] > mx)
    mx = Array[i]
  End if
  if(Array[i] < mn)
    mn = Array[i]
  End if
End for
Print mx - mn
```

A) 8 B) 9 C) 6 D) 4

Q7.

```
String s
Integer i, c
Set s = "SUCCESS", c = 0
for(each i from 0 to LENGTH(s) - 2)
  if(s[i] == s[i+1])
    c = c + 1
  End if
End for
Print c
```

A) 1 B) 2 C) 3 D) 4

Q8.

```
Integer a, b, c
Set a = 1, b = 2, c = 3
if(a < b AND b < c)
  Print a + c
else
  Print a - c
End if
```

A) -2 B) 4 C) 6 D) 3

Q9.

```
Integer n, s, i, j
Set n = 5, s = 0
for(each i from 1 to n)
  for(each j from 1 to i)
    s = s + 1
  End for
End for
Print s
```

A) 15 B) 25 C) 10 D) 5

Q10.

```
Integer Array[5], i, minIdx, t
Set Array = {2, 4, 1, 5, 3}
Set minIdx = 0
for(each i from 1 to 4)
  if(Array[i] < Array[minIdx])
    minIdx = i
  End if
End for
t = Array[0]
Array[0] = Array[minIdx]
Array[minIdx] = t
Print Array[0], Array[2]
```

A) 1 2 B) 2 1 C) 1 4 D) 2 4

Q11.

```
Integer x, i
Set x = 0
for(each i from 1 to 4)
  x = x * 10 + i
End for
Print x
```

A) 1234 B) 10 C) 4321 D) 4

Q12.

```
Integer a, b
Set a = 17, b = 5
a = a - (a MOD b)
Print a
```

A) 15 B) 12 C) 17 D) 2

Q13.

```
Integer p, q, r
Set p = 1, q = 2, r = 0
r = (p XOR q) OR (r OR 1)
Print r
```

A) 1 B) 2 C) 3 D) 0

Q14.

```
Integer n, c
Set n = 6, c = 1
while(n > 0)
  c = c * n
  n = n - 2
End while
Print c
```

A) 48 B) 720 C) 12 D) 24

Q15.

```
String s1, s2
Integer i, f
Set s1 = "abc", s2 = "cba", f = 1
for(each i from 0 to 2)
  if(s1[i] != s2[2 - i])
    f = 0
  End if
End for
Print f
```

A) 0 B) 1 C) 3 D) 2

Q16.

```
Integer i, j, c
Set c = 0, i = 1
while(i <= 4)
  Set j = 1
  while(j <= i)
    c = c + j
    j = j + 1
  End while
  i = i + 1
End while
Print c
```

A) 20 B) 10 C) 16 D) 24

Q17.

```
Integer Array[4], i, d, v
Set Array = {1, 0, 1, 1}
Set v = 0, d = 1
for(each i from 3 down to 0)
  v = v + Array[i] * d
  d = d * 2
End for
Print v
```

A) 11 B) 13 C) 7 D) 1011

Q18.

```
Integer n, s
Set n = 234, s = 0
while(n > 0)
  s = s * 10 + (n MOD 10)
  n = n / 10
End while
if(s > 234)
  Print "GREATER"
else
  Print "NOT GREATER"
End if
```

A) GREATER B) NOT GREATER C) 432 D) 234

Q19.

```
Integer i, c
Set c = 0
for(each i from 2 to 10)
  if(i MOD 2 == 0 OR i MOD 5 == 0)
    c = c + 1
  End if
End for
Print c
```

A) 5 B) 6 C) 7 D) 4

Q20.

```
Integer Array[5], i, j, t
Set Array = {9, 7, 5, 3, 1}
for(each i from 0 to 1)
  for(each j from 0 to 3 - i)
    if(Array[j] > Array[j+1])
      t = Array[j]
      Array[j] = Array[j+1]
      Array[j+1] = t
    End if
  End for
End for
Print Array[0], Array[4]
```

A) 5 9 B) 1 9 C) 9 1 D) 3 9

Section A - Computer Fundamentals (Q21 to Q30)

Q21. In C, given int a = 5; int b = a++; int c = ++a; what are b and c?

A) 5 6 B) 5 7 C) 6 7 D) 6 6

Q22. Which sort is stable with O(n log n) worst case?

A) QuickSort B) Merge sort C) Heap sort D) Selection sort

Q23. Which Java collection disallows duplicates?

A) ArrayList B) LinkedList C) HashSet D) Vector

Q24. SQL DISTINCT does what?

A) Sorts rows B) Removes duplicate rows from the result

C) Joins tables D) Filters groups

Q25. Best case time complexity of insertion sort?

A) O(1) B) O(n) C) O(n log n) D) O(n squared)

Q26. Where are non-static local variables stored in C?

A) Heap B) Stack C) Data segment D) Text segment

Q27. ARP resolves:

A) Domain name to IP B) IP to MAC C) MAC to IP D) Port to process

Q28. Which gives O(1) average lookup by key?

A) Array B) Linked list C) Hash table D) BST

Q29. An abstract class:

A) Can be instantiated directly

B) Cannot be instantiated and may contain abstract methods

C) Must contain only abstract methods D) Cannot have constructors

Q30. Reading data written by an uncommitted transaction is a:

A) Phantom read B) Dirty read C) Lost update D) Non-repeatable read

Section B - English Communication (Q31 to Q40)

Q31. Each of the modules ____ its own test suite.

A) have B) has C) are having D) were having

Q32. Synonym of ROBUST:

A) fragile B) resilient C) sluggish D) minimal

Q33. Antonym of REDUNDANT:

A) superfluous B) essential C) excessive D) repetitive

Q34. The outage was attributed ____ a misconfigured load balancer.

A) on B) to C) for D) with

Q35. Find the error: "Neither of the two approaches are viable."

A) Neither of B) the two approaches C) are viable D) No error

Q36. Passive of "The architect will review the design tomorrow."

A) The design will be reviewed by the architect tomorrow

B) The design is reviewed by the architect tomorrow

C) The design was reviewed by the architect tomorrow

D) The design would be reviewed by the architect tomorrow

Q37. "Low-hanging fruit" means:

A) Easy wins B) Unripe work C) Low priority D) Cheap tools

Q38. Rearrange: (P) after the schema change (Q) the reports (R) were regenerated (S) had been applied

A) PSQR B) QRPS C) PQRS D) QPRS

Q39. Indirect speech: She said, "I will send the report tomorrow."

A) She said that she will send the report tomorrow

B) She said that she would send the report the next day

C) She said that she sent the report tomorrow

D) She says she would send the report tomorrow

Q40. Read the passage and answer.

```
Technical debt is not a defect. It is a deliberate trade-off taken to ship
sooner, and it becomes a problem only when teams stop tracking what they
borrowed.
```

Technical debt turns harmful when:

A) It is incurred at all B) It goes unrecorded

C) Releases are delayed D) Code is refactored

Section C - Game-Based Cognitive (Q41 to Q44)

Q41. GeoSudo. What replaces the question mark?

```
CIR   PLS   SQR   TRI
PLS   SQR   TRI   CIR
SQR   TRI   CIR   PLS
TRI   CIR   PLS    ?
```

A) CIR B) TRI C) SQR D) PLS

Q42. digitChallenge. Place 3, 4 and 6 into the blanks so the equation holds. Each digit is used once.

```
_  +  _  *  _  =  27
```

A) 3, 4, 6 B) 4, 3, 6 C) 6, 4, 3 D) 4, 6, 3

Q43. Inductive reasoning. What is the next term?

```
2,  5,  10,  17,  26,  ?
```

A) 35 B) 36 C) 37 D) 38

Q44. motionChallenge, discrete tick. A 5 by 5 grid. The ball starts at row 1, column 1 and the hole is at row 5, column 5. One orthogonal move per step. Cells (1,3), (2,3), (3,3) and (4,3) are blocked. Minimum number of steps?

A) 8 B) 9 C) 10 D) 12

Section D - Coding Round (45 minutes)

Problem 1 - Anagram Check

Given two strings, determine whether they are anagrams of each other. Case-sensitive. Return true or false. O(n) time.

```
Input   "listen", "silent"      Output   true
Input   "hello", "world"        Output   false
```

Problem 2 - Balanced Parentheses

Given a string containing only the six bracket characters, determine whether every bracket is closed by the correct type in the correct order. Use a stack. O(n) time.

```
Input   "{[()]}"      Output   true
Input   "{[(])}"      Output   false
Input   "((("         Output   false
```

Test Set 5

Standard Analyst band.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 25, b = 7
c = (a / b) * b + (a MOD b)
Print c
```

A) 21 B) 25 C) 4 D) 28

Q2.

```
Integer p, q
Set p = 1, q = 6
p = p << q
p = p >> 3
Print p
```

A) 8 B) 64 C) 16 D) 2

Q3.

```
Integer i, s
Set s = 0
for(each i from 1 to 9)
  if(i MOD 2 == 1)
    s = s + i
  End if
End for
Print s
```

A) 20 B) 25 C) 45 D) 24

Q4.

```
Integer fun(Integer n)
  if(n <= 0)
    return 0
  End if
  return n + fun(n - 3)
End function fun()
Print fun(10)
```

A) 22 B) 25 C) 18 D) 10

Q5.

```
Integer n, c
Set n = 3125, c = 0
while(n MOD 5 == 0)
  n = n / 5
  c = c + 1
End while
Print c
```

A) 4 B) 5 C) 6 D) 3

Q6.

```
Integer Array[7], lo, hi, mid, key, c
Set Array = {1, 3, 5, 7, 9, 11, 13}
Set lo = 0, hi = 6, key = 4, c = 0
while(lo <= hi)
  mid = (lo + hi) / 2
  c = c + 1
  if(Array[mid] == key)
    break
  else if(Array[mid] < key)
    lo = mid + 1
  else
    hi = mid - 1
  End if
End while
Print c
```

A) 2 B) 3 C) 4 D) 7

Q7.

```
String s
Integer i, c
Set s = "MISSISSIPPI", c = 0
for(each i from 0 to LENGTH(s) - 1)
  if(s[i] == 'S')
    c = c + 1
  End if
End for
Print c
```

A) 2 B) 3 C) 4 D) 5

Q8.

```
Integer a, b, c
Set a = 2, b = 10, c = 0
while(a < b)
  a = a * 2
  b = b + 1
  c = c + 1
End while
Print c
```

A) 2 B) 3 C) 4 D) Infinite loop

Q9.

```
Integer x, y
Set x = 8, y = 3
if(x MOD y == 0)
  Print "DIVISIBLE"
else if(x > y * 2)
  Print "LARGE"
else
  Print "SMALL"
End if
```

A) DIVISIBLE B) LARGE C) SMALL D) No output

Q10.

```
Integer n, s, i
Set n = 4, s = 0
for(each i from 1 to n)
  s = s + (n - i + 1)
End for
Print s
```

A) 10 B) 4 C) 16 D) 6

Q11.

```
Integer Array[6], i, x
Set Array = {4, 4, 2, 2, 6, 6}
Set x = 0
for(each i from 0 to 5)
  x = x XOR Array[i]
End for
Print x
```

A) 0 B) 4 C) 24 D) 6

Q12.

```
Integer a, b, c
Set a = 7, b = 7
if(a == b)
  c = a * b
else
  c = a + b
End if
Print c MOD 10
```

A) 4 B) 9 C) 49 D) 14

Q13.

```
Integer i, j, s
Set s = 0
for(each i from 1 to 3)
  for(each j from 3 down to 1)
    if(i + j == 4)
      s = s + 1
    End if
  End for
End for
Print s
```

A) 2 B) 3 C) 4 D) 9

Q14.

```
Integer n, r
Set n = 1234, r = 0
while(n > 0)
  if((n MOD 10) MOD 2 == 0)
    r = r + (n MOD 10)
  End if
  n = n / 10
End while
Print r
```

A) 4 B) 6 C) 10 D) 2

Q15.

```
String s
Integer i
Set s = "ABCD"
for(each i from 0 to LENGTH(s) - 1)
  if(i MOD 2 == 0)
    Print s[i]
  End if
End for
```

A) AC B) BD C) ABCD D) DCBA

Q16.

```
Integer a, b, t
Set a = 48, b = 18
while(b != 0)
  t = b
  b = a MOD b
  a = t
End while
Print a
```

A) 6 B) 12 C) 18 D) 3

Q17.

```
Integer Array[5], i, s
Set Array = {1, 2, 3, 4, 5}
Set s = 0
for(each i from 0 to 4)
  if(Array[i] MOD 2 == 0)
    s = s + Array[i]
  else
    s = s - Array[i]
  End if
End for
Print s
```

A) 3 B) -3 C) 15 D) 6

Q18.

```
Integer p, q, r, i
Set p = 10, q = 3, r = 1
for(each i from 1 to q)
  r = r * p
End for
Print r / 100
```

A) 10 B) 100 C) 1000 D) 1

Q19.

```
Integer n, c
Set n = 255, c = 0
while(n > 0)
  c = c + (n AND 1)
  n = n >> 1
End while
Print c
```

A) 7 B) 8 C) 255 D) 1

Q20.

```
Integer Array[5], i, j, c
Set Array = {1, 2, 3, 4, 5}
Set c = 0
for(each i from 0 to 4)
  for(each j from i + 1 to 4)
    if(Array[i] + Array[j] == 6)
      c = c + 1
    End if
  End for
End for
Print c
```

A) 1 B) 2 C) 3 D) 4

Section A - Computer Fundamentals (Q21 to Q30)

Q21. Inserting at the head of a singly linked list costs:

A) O(1) B) O(n) C) O(log n) D) O(n squared)

Q22. Which C++ feature lets an operator work with user-defined types?

A) Templates B) Operator overloading C) Friend functions D) Virtual inheritance

Q23. In Java, if a class declares no constructor:

A) Compile error B) The compiler supplies a no-argument default constructor

C) The class cannot be instantiated D) Runtime error

Q24. Which ignores NULL values?

A) COUNT(*) B) COUNT(column) C) Both D) Neither

Q25. Height of a balanced BST with n nodes?

A) O(n) B) O(log n) C) O(1) D) O(n log n)

Q26. Which page replacement algorithm can exhibit Belady's anomaly?

A) LRU B) FIFO C) Optimal D) LFU

Q27. DNS uses which transport protocol for standard queries?

A) TCP B) UDP C) ICMP D) ARP

Q28. Which structure implements recursion internally?

A) Queue B) Stack C) Heap D) Graph

Q29. In C, malloc returns:

A) An int B) A void pointer C) A char pointer D) The allocated size

Q30. Which dependency does 2NF eliminate?

A) Transitive B) Partial C) Multivalued D) Join

Section B - English Communication (Q31 to Q40)

Q31. A set of guidelines ____ been published for the new API.

A) have B) has C) are D) were

Q32. Synonym of MITIGATE:

A) worsen B) alleviate C) ignore D) accelerate

Q33. Antonym of VOLATILE:

A) unstable B) stable C) erratic D) explosive

Q34. The team is averse ____ changing the release date.

A) of B) to C) for D) with

Q35. Find the error: "He is one of the engineers who has contributed to the framework."

A) He is one of B) the engineers C) who has contributed D) No error

Q36. Passive of "They are testing the new build."

A) The new build is being tested B) The new build was tested

C) The new build has been tested D) The new build is tested

Q37. "Move the needle" means:

A) Make a noticeable difference B) Adjust settings

C) Delay D) Measure precisely

Q38. Rearrange: (P) the release notes (Q) once the build passes (R) will be published (S) regression testing

A) QSPR B) PQRS C) QPRS D) PRQS

Q39. Indirect speech: The manager asked, "Have you closed the ticket?"

A) The manager asked if I had closed the ticket

B) The manager asked have you closed the ticket

C) The manager asked if I have closed the ticket

D) The manager asks if I closed the ticket

Q40. Read the passage and answer.

```
Adding engineers to a late project rarely speeds it up. New members consume
the time of those already productive, and communication paths grow faster
than headcount.
```

The central claim rests on:

A) New engineers lacking skill B) Coordination cost rising non-linearly

C) Budget constraints D) Poor documentation

Section C - Game-Based Cognitive (Q41 to Q44)

Q41. GeoSudo. What replaces the question mark?

```
SQR   TRI   CIR   PLS
TRI   CIR   PLS   SQR
CIR   PLS   SQR   TRI
PLS   SQR   TRI    ?
```

A) CIR B) TRI C) SQR D) PLS

Q42. digitChallenge. Insert operators so the equation holds.

```
20  _  5  _  3  =  1
```

A) / then - B) - then / C) / then + D) * then -

Q43. Inductive reasoning. What is the next term?

```
1,  2,  6,  24,  120,  ?
```

A) 600 B) 620 C) 720 D) 840

Q44. Rule application. Button P adds 7. Button Q multiplies by 2. Button R subtracts 4. Starting at 3, apply Q, P, Q, R in that order. Final value?

A) 18 B) 20 C) 22 D) 26

Section D - Coding Round (45 minutes)

Problem 1 - Missing Number

An array holds n distinct integers drawn from the range 1 to n+1, with exactly one value missing. Return the missing value. O(n) time, O(1) space. Do not sort.

```
Input   [1, 2, 4, 5, 6]     Output   3
Input   [2, 3, 4]           Output   1
```

Problem 2 - Rotate Matrix 90 Degrees Clockwise

Given an n by n integer matrix, rotate it 90 degrees clockwise in place. No second matrix is allowed.

```
Input        Output
1 2 3        7 4 1
4 5 6        8 5 2
7 8 9        9 6 3
```

Part Two - Senior Analyst Band

Sets H1 to H5 sit deliberately above Sets 1 to 5. They target the Senior Analyst gate, the track carrying the higher package, and they draw on multi-dimensional arrays, recursion with non-obvious base cases, bit manipulation, two-pointer partitioning, pass-by-value semantics, and layered switchChallenge operators.

On these five sets, 55 to 65 percent on Section A is a solid result for a well-prepared candidate. Scoring above 80 percent means you are comfortably past what the live assessment demands and the remaining gain is in speed rather than content.

Section C now uses the full Aon smartPredict format set: switchChallenge at one, two and three operator layers, digitChallenge with the digit-distinctness rule, gridChallenge with memory rounds separated by spatial distractors, and motionChallenge as discrete-tick reflection tracking. Review the notation reference at the front of this book before attempting them.

Timing per set: Section A 30 minutes, Section B 10 minutes, Section C 10 minutes, Section D 45 minutes.

Test Set H1

Senior Analyst band - medium to hard.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 2, b = 3, c = 4
a = c / b * a + b MOD a
Print a
```

A) 4 B) 2 C) 3 D) 5

Q2.

```
Integer fun(Integer n)
  if(n == 0)
    return 0
  End if
  Print n MOD 2
  return fun(n / 2)
End function fun()
Call fun(13)
```

A) 1011 B) 1101 C) 1010 D) 0111

Q3.

```
Integer i, c
Set c = 0, i = 1
while(i < 30)
  if(i MOD 3 == 0)
    i = i + 5
  else
    i = i + 2
  End if
  c = c + 1
End while
Print i, c
```

A) 28 9 B) 30 10 C) 30 9 D) 31 10

Q4.

```
Integer M[3][3], i, s
Set M = {{1,2,3},{4,5,6},{7,8,9}}
Set s = 0
for(each i from 0 to 2)
  s = s + M[i][2 - i] - M[i][i]
End for
Print s
```

A) 15 B) 30 C) 4 D) 0

Q5.

```
Integer a, b, c
Set a = 4, b = 0, c = 0
while(b < 3 AND a > b)
  b = b + 1
  a = a - 1
  c = c + a + b
End while
Print c
```

A) 12 B) 8 C) 4 D) 10

Q6.

```
Integer modify(Integer x)
  x = x * 2
  return x + 1
End function modify()
Integer n, r
Set n = 5
r = modify(n)
Print n, r
```

A) 10 11 B) 5 10 C) 5 11 D) 10 21

Q7.

```
String s
Integer i, c
Set s = "ABCDEFG", c = 0
for(each i from 0 to LENGTH(s) - 1)
  if((i + s[i] - 'A') MOD 3 == 0)
    c = c + 1
  End if
End for
Print c
```

A) 2 B) 3 C) 4 D) 7

Q8.

```
Integer A[4][4], i, j, f, r
Set A = {{1,2,3,4},{5,6,7,8},{9,10,11,12},{13,14,15,16}}
Set f = 0, r = -1
for(each i from 0 to 3)
  for(each j from 0 to 3)
    if(A[i][j] MOD 7 == 0)
      r = i * 4 + j
      f = 1
      break
    End if
  End for
  if(f == 1)
    break
  End if
End for
Print r
```

A) 6 B) 7 C) 14 D) 5

Q9.

```
Integer f(Integer n)
  if(n <= 2)
    return 1
  End if
  return f(n - 1) + f(n - 3)
End function f()
Print f(7)
```

A) 6 B) 7 C) 9 D) 13

Q10.

```
Integer n, m, c
Set n = 45, m = 0, c = 0
while(n > 0)
  if((n AND 1) == 1)
    m = m OR (1 << c)
  End if
  n = n >> 1
  c = c + 1
End while
Print m, c
```

A) 45 5 B) 45 6 C) 90 6 D) 0 6

Q11.

```
Integer n, s
Set n = 0, s = 0
repeat
  n = n + 3
  s = s + n
until(s > 20)
Print n, s
```

A) 9 18 B) 12 18 C) 15 45 D) 12 30

Q12.

```
Integer A[5], i
Set A = {2, 3, 1, 5, 4}
for(each i from 1 to 4)
  A[i] = A[i] + A[i-1]
End for
Print A[2], A[4]
```

A) 6 15 B) 4 15 C) 6 9 D) 1 4

Q13.

```
Integer A[4], i, j
Set A = {1, 2, 3, 4}
for(each i from 0 to 3)
  j = 3 - i
  A[i] = A[i] XOR A[j]
  A[j] = A[i] XOR A[j]
  A[i] = A[i] XOR A[j]
End for
Print A[0], A[1], A[2], A[3]
```

A) 4 3 2 1 B) 1 2 3 4 C) 4 2 3 1 D) 0 0 0 0

Q14.

```
Integer x, y, r
Set x = 7, y = 12
if(x MOD 2 == 1 AND y MOD 2 == 0)
  r = x * y
else if(x + y > 15)
  r = x + y
else
  r = y - x
End if
Print r MOD 11
```

A) 5 B) 8 C) 7 D) 4

Q15.

```
Integer i, j, c
Set c = 0
for(each i from 1 to 4)
  for(each j from 1 to 4)
    if((i * j) MOD 2 == 0 AND i != j)
      c = c + 1
    End if
  End for
End for
Print c
```

A) 8 B) 10 C) 12 D) 6

Q16.

```
Integer n, r, d
Set n = 897, r = 0
while(n > 0)
  d = n MOD 10
  if(d MOD 3 == 0)
    r = r * 10 + (d / 3)
  else
    r = r * 10 + d
  End if
  n = n / 10
End while
Print r
```

A) 798 B) 837 C) 873 D) 738

Q17.

```
Integer A[6], i, j, t
Set A = {3, 8, 2, 9, 1, 7}
Set i = 0, j = 5
while(i < j)
  while(i < j AND A[i] MOD 2 == 1)
    i = i + 1
  End while
  while(i < j AND A[j] MOD 2 == 0)
    j = j - 1
  End while
  if(i < j)
    t = A[i]
    A[i] = A[j]
    A[j] = t
  End if
End while
Print A[0], A[1], A[2]
```

A) 3 7 1 B) 3 8 2 C) 3 7 9 D) 1 7 3

Q18.

```
String f(Integer n)
  if(n == 0)
    return ""
  End if
  return f(n / 2) + (n MOD 2)
End function f()
Print f(11)
```

A) 1101 B) 1011 C) 11 D) 1010

Q19.

```
Integer M[3][3], i, j, t, s
Set M = {{1,2,3},{4,5,6},{7,8,9}}
for(each i from 0 to 2)
  for(each j from i + 1 to 2)
    t = M[i][j]
    M[i][j] = M[j][i]
    M[j][i] = t
  End for
End for
Set s = 0
for(each j from 0 to 2)
  s = s + M[0][j]
End for
Print s
```

A) 6 B) 18 C) 12 D) 15

Q20.

```
Integer n, i, c
Set n = 100, c = 0, i = 2
while(i * i <= n)
  if(n MOD i == 0)
    c = c + 1
    while(n MOD i == 0)
      n = n / i
    End while
  End if
  i = i + 1
End while
if(n > 1)
  c = c + 1
End if
Print c
```

A) 3 B) 2 C) 4 D) 1

Section A - Computer Fundamentals (Q21 to Q30)

Q21. An outer loop runs i from 1 to n. Inside it, j is reset to 1 and a while loop doubles j until j is no longer less than n. Total complexity?

A) O(n) B) O(n log n) C) O(n squared) D) O(log n)

Q22. In C++, deleting a derived object through a base pointer when the base destructor is not virtual causes:

A) A compile error

B) Undefined behaviour - the derived destructor is not invoked

C) A leak confined to the base part D) Nothing

Q23. Which C++ mechanism resolves the diamond problem?

A) Friend classes B) Templates C) Virtual inheritance D) Namespaces

Q24. Which isolation level permits phantom reads but prevents dirty and non-repeatable reads?

A) Read Uncommitted B) Read Committed C) Repeatable Read D) Serializable

Q25. Hash table with separate chaining, n keys and m buckets. Average search cost is:

A) O(1) unconditionally B) O(n) C) O(1 + n/m) D) O(log n)

Q26. Breaking which single condition guarantees no deadlock?

A) Mutual exclusion B) Circular wait C) Bounded buffer D) Fairness

Q27. T(n) = T(n-1) + O(1) with T(1) = O(1) describes:

A) O(log n) B) O(n) C) O(n log n) D) O(2 to the power n)

Q28. In a B+ tree, where do the data records or pointers to them reside?

A) All nodes equally B) Internal nodes only C) Leaf nodes only D) The root

Q29. The volatile keyword in C tells the compiler:

A) The variable is constant B) Not to optimise away reads of the variable

C) To place it on the heap D) To make access atomic

Q30. Which traversal, paired with inorder, reconstructs a binary tree uniquely?

A) Level order only B) Preorder or postorder C) Neither D) Inorder repeated

Section B - English Communication (Q31 to Q38)

Q31. The committee, whose members ____ divided on the proposal, ____ yet to issue a recommendation.

A) is / are B) are / has C) are / is D) is / is

Q32. Synonym of PERFUNCTORY:

A) thorough B) cursory C) enthusiastic D) deliberate

Q33. Antonym of INTRANSIGENT:

A) stubborn B) accommodating C) rigid D) hostile

Q34. "Hoist with one's own petard" means:

A) Harmed by one's own scheme B) Promoted unexpectedly

C) Raising a flag D) Abandoning a plan

Q35. Find the error: "Hardly had the deployment finished than the alerts began firing."

A) Hardly had B) the deployment finished C) than the alerts D) No error

Q36. Which sentence has correct parallel structure?

A) The tool validates input, logs errors, and it retries failed calls

B) The tool validates input, logs errors, and retries failed calls

C) The tool validates input, error logging, and retrying failed calls

D) The tool is validating input, logs errors, and retried failed calls

Q37. Rearrange: (P) had the rollback been scripted (Q) the outage (R) by more than an hour (S) would have been shortened

A) PQSR B) QSRP C) PQRS D) QPSR

Q38. Read the passage and answer.

```
Advocates of formal verification note that testing can only demonstrate the
presence of defects, never their absence. Critics reply that a proof
establishes correctness relative to a specification, and specifications are
themselves written by humans. Neither camp disputes the other's premise.
```

The final sentence implies the disagreement concerns:

A) Whether testing finds defects

B) Where the residual risk sits, not whether it exists

C) The cost of verification D) The accuracy of proofs

Section C - Game-Based Cognitive (Q39 to Q44)

Q39. switchChallenge, single operator. The operator's k-th digit names which input position supplies output position k. Find the operator.

```
Input     STR   CIR   TRI   SQR
Output    TRI   STR   SQR   CIR
```

A) 2413 B) 3142 C) 3124 D) 4132

Q40. switchChallenge, two layers. Operator 1 is given. Find the missing Operator 2.

```
Input         STR   CIR   TRI   SQR
Operator 1    2431
Intermediate  (hidden)
Operator 2    ?
Output        STR   TRI   CIR   SQR
```

A) 3421 B) 4132 C) 4312 D) 2143

Q41. switchChallenge, three layers. Operators 1 and 3 are given. Find the missing middle operator.

```
Input         STR   CIR   TRI   SQR
Operator 1    3142
Operator 2    ?
Operator 3    2143
Output        STR   CIR   SQR   TRI
```

A) 4213 B) 2431 C) 4123 D) 1342

Q42. digitChallenge. Fill the blanks with digits 1 to 9. Each digit may be used only once in the equation. Standard precedence applies.

```
_  *  _  -  _  /  _  =  20
```

A) 4, 6, 8, 2 B) 7, 3, 2, 2 C) 3, 8, 6, 2 D) 9, 2, 4, 1

Q43. gridChallenge. Work the three stages in order and do not look ahead.

Stage 1 - memorise this grid for 10 seconds, then cover it.

```
.   X   .   .
.   .   .   X
X   .   .   .
.   .   X   .
```

Stage 2 - spatial distractor. Are these two grids mirror images of each other across the vertical centre line?

```
Left            Right
X   .   X       X   .   X
.   X   .       .   X   .
X   .   .       .   .   X
```

Stage 3 - recall. In the Stage 1 grid, how many dots lay in the bottom two rows?

Q44. motionChallenge, discrete tick. A ball sits at row 1, column 1 of a 5 by 5 grid, moving diagonally down and to the right one cell per tick. It reflects off any wall it strikes, and on a corner both components reverse at once. Where is it after 6 ticks?

A) row 3, col 3 B) row 5, col 5 C) row 4, col 4 D) row 2, col 2

Section D - Coding Round (45 minutes, C / C++ / Java)

Problem 1 - Longest Subarray With Sum K

Given an integer array that may contain negatives and zeros, return the LENGTH of the longest contiguous subarray whose elements sum to exactly K. Return 0 if none exists. Use prefix sums with a hash map for O(n); the O(n squared) solution will time out.

```
Input   arr = [10, 5, 2, 7, 1, 9], K = 15     Output   4    (5, 2, 7, 1)
Input   arr = [-1, 2, 3, -2, 4], K = 4        Output   5
```

Problem 2 - Next Greater Element

For every element of an array, output the first element to its right that is strictly greater. Output -1 where none exists. O(n) using a monotonic stack.

```
Input   [4, 5, 2, 25]        Output   [5, 25, 25, -1]
Input   [13, 7, 6, 12]       Output   [-1, 12, 12, -1]
```

Test Set H2

Senior Analyst band - medium to hard.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 10, b = 4, c = 3
a = a - b * c / b + c
Print a
```

A) 1 B) 4 C) 10 D) 13

Q2.

```
Integer f(Integer n, Integer d)
  if(n == 0)
    return 0
  End if
  return (n MOD d) + f(n / d, d)
End function f()
Print f(255, 4)
```

A) 9 B) 3 C) 15 D) 12

Q3.

```
Integer i, j, c
Set c = 0, i = 1, j = 10
while(i < j)
  if((i + j) MOD 3 == 0)
    i = i + 2
  else
    j = j - 1
  End if
  c = c + 1
End while
Print i, j, c
```

A) 5 6 6 B) 5 5 7 C) 3 5 7 D) 5 5 8

Q4.

```
Integer M[3][4], i, j, s
Set M = {{2,4,6,8},{1,3,5,7},{9,7,5,3}}
Set s = 0
for(each i from 0 to 2)
  for(each j from 0 to 3)
    if((i + j) MOD 2 == 0)
      s = s + M[i][j]
    else
      s = s - M[i][j]
    End if
  End for
End for
Print s
```

A) -4 B) 0 C) 4 D) 12

Q5.

```
Integer swap(Integer p, Integer q)
  Integer t
  t = p
  p = q
  q = t
  return p + q
End function swap()
Integer x, y, r
Set x = 3, y = 8
r = swap(x, y)
Print x, y, r
```

A) 8 3 11 B) 3 8 0 C) 8 3 5 D) 3 8 11

Q6.

```
String s
Integer i, c
Set s = "AABBBCC", c = 1
for(each i from 1 to LENGTH(s) - 1)
  if(s[i] != s[i-1])
    c = c + 1
  End if
End for
Print c
```

A) 2 B) 3 C) 4 D) 7

Q7.

```
Integer A[6], i, mx, cur
Set A = {2, -3, 4, -1, 2, -5}
Set mx = A[0], cur = A[0]
for(each i from 1 to 5)
  if(cur + A[i] > A[i])
    cur = cur + A[i]
  else
    cur = A[i]
  End if
  if(cur > mx)
    mx = cur
  End if
End for
Print mx
```

A) 2 B) 4 C) 5 D) 6

Q8.

```
Integer n, c, t
Set n = 6, c = 0
for(each t from 1 to n)
  if(n MOD t == 0)
    c = c + t
  End if
End for
if(c - n == n)
  Print "PERFECT"
else
  Print c - n
End if
```

A) 6 B) 12 C) PERFECT D) 0

Q9.

```
Integer f(Integer n)
  Integer c
  Set c = 0
  while(n != 1 AND c < 10)
    if(n MOD 2 == 0)
      n = n / 2
    else
      n = 3 * n + 1
    End if
    c = c + 1
  End while
  return c
End function f()
Print f(6)
```

A) 7 B) 8 C) 9 D) 10

Q10.

```
Integer a, b
Set a = 29, b = 0
while(a > 0)
  b = b * 2 + (a MOD 2)
  a = a / 2
End while
Print b
```

A) 15 B) 29 C) 31 D) 23

Q11.

```
Integer A[5], i, j, c
Set A = {1, 3, 2, 3, 1}
Set c = 0
for(each i from 0 to 4)
  for(each j from 0 to 4)
    if(A[i] == A[j] AND i < j)
      c = c + 1
    End if
  End for
End for
Print c
```

A) 1 B) 2 C) 3 D) 4

Q12.

```
Integer x, y, z
Set x = 5
y = x AND (x - 1)
z = x OR (x + 1)
Print y, z
```

A) 5 6 B) 1 7 C) 4 5 D) 4 7

Q13.

```
Integer A[4][4], i, j, s
Set A = {{1,2,3,4},{5,6,7,8},{9,1,2,3},{4,5,6,7}}
Set s = 0
for(each i from 0 to 3)
  for(each j from 0 to 3)
    if(i > j)
      s = s + A[i][j]
    End if
  End for
End for
Print s
```

A) 16 B) 25 C) 30 D) 46

Q14.

```
Integer i, s
Set s = 0, i = 1
while(TRUE)
  s = s + i
  if(s > 15)
    break
  End if
  i = i + 2
End while
Print i, s
```

A) 9 25 B) 7 9 C) 5 16 D) 7 16

Q15.

```
String s1, s2
Integer i, c
Set s1 = "ABCD", s2 = "CDAB", c = 0
for(each i from 0 to 3)
  if(s1[i] == s2[(i + 2) MOD 4])
    c = c + 1
  End if
End for
Print c
```

A) 0 B) 2 C) 3 D) 4

Q16.

```
Integer f(Integer n)
  if(n < 10)
    return n
  End if
  return f(n / 10 + n MOD 10)
End function f()
Print f(789)
```

A) 3 B) 6 C) 9 D) 24

Q17.

```
Integer A[7], lo, hi, mid, key
Set A = {1, 2, 4, 4, 4, 7, 9}
Set lo = 0, hi = 6, key = 4
while(lo < hi)
  mid = (lo + hi) / 2
  if(A[mid] < key)
    lo = mid + 1
  else
    hi = mid
  End if
End while
Print lo
```

A) 2 B) 3 C) 4 D) 5

Q18.

```
Integer A[5], i, t, n
Set A = {4, 2, 7, 1, 9}
Set n = 5
for(each i from 0 to n / 2 - 1)
  t = A[i]
  A[i] = A[n - 1 - i]
  A[n - 1 - i] = t
End for
Print A[0], A[2], A[4]
```

A) 4 7 9 B) 9 2 4 C) 9 7 4 D) 1 7 2

Q19.

```
Integer n, c, p
Set n = 12, c = 0, p = 1
while(p <= n)
  if((n AND p) != 0)
    c = c + 1
  End if
  p = p << 1
End while
Print c
```

A) 1 B) 2 C) 3 D) 4

Q20.

```
Integer A[6], i, w
Set A = {0, 1, 0, 3, 0, 5}
Set w = 0
for(each i from 0 to 5)
  if(A[i] != 0)
    A[w] = A[i]
    w = w + 1
  End if
End for
while(w <= 5)
  A[w] = 0
  w = w + 1
End while
Print A[0], A[1], A[2], A[3]
```

A) 0 1 0 3 B) 1 3 5 3 C) 5 3 1 0 D) 1 3 5 0

Section A - Computer Fundamentals (Q21 to Q30)

Q21. T(n) = 2T(n/2) + n resolves by the Master Theorem to:

A) O(n) B) O(n log n) C) O(n squared) D) O(log n)

Q22. How many null keys does a Java HashMap permit?

A) 0 B) 1 C) Unlimited D) Compile error

Q23. Which C++ idiom guarantees resource release on scope exit, including during exception unwinding?

A) goto cleanup B) RAII via destructors

C) Static storage D) Preprocessor macros

Q24. A relation is in BCNF when, for every non-trivial functional dependency X to Y:

A) Y is a candidate key B) X is a superkey

C) X is a prime attribute D) Y is prime

Q25. Which is true of a min-heap holding n elements?

A) The largest element is at the root

B) Searching an arbitrary key is O(log n)

C) Extract-min is O(log n) D) It is a sorted array

Q26. TCP delivers reliability chiefly through:

A) Checksums alone B) Routing tables

C) Sequence numbers plus acknowledgements with retransmission

D) Port multiplexing

Q27. Inserting already-sorted input into an unbalanced BST costs, in the worst case:

A) O(log n) B) O(1) C) O(n log n) D) O(n) per insertion

Q28. For int *p = arr; the expression p + 3 advances the address by:

A) 3 bytes B) 1 byte C) 3 times sizeof(int) bytes D) sizeof(p) bytes

Q29. Which index structure best serves range queries on an ordered column?

A) Hash index B) B+ tree index C) Bitmap index D) Inverted index

Q30. Two threads increment a shared counter without synchronisation. The defect class is:

A) Deadlock B) Starvation C) Race condition D) Livelock

Section B - English Communication (Q31 to Q38)

Q31. Not one of the proposed mitigations ____ the underlying race condition, though several ____ its symptoms.

A) address / masks B) addresses / mask

C) address / mask D) addresses / masks

Q32. Synonym of TENUOUS:

A) robust B) flimsy C) explicit D) permanent

Q33. Antonym of PROLIFERATE:

A) multiply B) dwindle C) spread D) escalate

Q34. "To paper over the cracks" means:

A) Conceal a problem rather than fix it B) Document thoroughly

C) Reinforce a design D) Finish a report

Q35. Find the error: "Neither the logs nor the metrics was conclusive, which left the team guessing."

A) Neither the logs B) nor the metrics C) was conclusive D) No error

Q36. Which sentence contains no dangling modifier?

A) Having reviewed the schema, the migration was approved

B) Having reviewed the schema, the architect approved the migration

C) Having reviewed the schema, approval was given

D) Reviewing the schema, the migration seemed sound

Q37. Rearrange: (P) no amount of caching (Q) if the query itself is unindexed (R) will make (S) the endpoint responsive

A) QPRS B) PRSQ C) PQRS D) QRPS

Q38. Read the passage and answer.

```
Estimation error in software is not symmetric. Tasks that finish early save
the hours they save; tasks that overrun consume not only their own overrun
but the slack of everything scheduled behind them. A plan built on average
estimates is therefore optimistic even when the averages are correct.
```

The argument turns on:

A) Estimates being systematically too low

B) Overruns propagating while early finishes do not

C) Averages being miscalculated D) Developers padding estimates

Section C - Game-Based Cognitive (Q39 to Q44)

Q39. switchChallenge, single operator.

```
Input     DIA   TRI   CIR   SQR
Output    SQR   CIR   DIA   TRI
```

A) 3421 B) 4321 C) 4312 D) 2431

Q40. switchChallenge, two layers. Find Operator 2. Note that the output equals the input, so Operator 2 must be the inverse of Operator 1.

```
Input         DIA   TRI   CIR   SQR
Operator 1    3142
Operator 2    ?
Output        DIA   TRI   CIR   SQR
```

A) 4231 B) 2431 C) 3142 D) 2413

Q41. switchChallenge, three layers. Find the middle operator.

```
Input         DIA   TRI   CIR   SQR
Operator 1    2143
Operator 2    ?
Operator 3    3412
Output        CIR   SQR   DIA   TRI
```

A) 1234 B) 2143 C) 4321 D) 3142

Q42. digitChallenge. Digits 1 to 9, each used only once in the equation.

```
_  *  _  +  _  /  _  =  31
```

A) 4, 7, 6, 2 B) 5, 6, 2, 2 C) 3, 9, 8, 4 D) 6, 5, 2, 1

Q43. gridChallenge. Work the three stages in order.

Stage 1 - memorise this grid for 10 seconds, then cover it.

```
.   .   X   .
X   .   .   .
.   .   .   X
.   X   .   .
```

Stage 2 - distractor. Are these two grids mirror images across the vertical centre line?

```
Left            Right
X   X   .       .   X   X
.   .   X       X   .   .
X   .   X       X   .   X
```

Stage 3 - recall. In the Stage 1 grid, how many dots lay strictly above the main diagonal, that is with column index greater than row index?

Q44. motionChallenge, discrete tick. A ball starts at row 2, column 1 of a 4 by 4 grid, moving diagonally down and to the right, reflecting off walls. Position after 5 ticks?

A) row 1, col 2 B) row 2, col 3 C) row 3, col 4 D) row 4, col 3

Section D - Coding Round (45 minutes)

Problem 1 - Majority Element

An array of length n contains an element appearing strictly more than n/2 times. Return it. O(n) time and O(1) extra space; hash-map solutions violate the space constraint.

```
Input   [2, 2, 1, 1, 1, 2, 2]     Output   2
Input   [3, 3, 4]                 Output   3
```

Problem 2 - Merge Overlapping Intervals

Given a list of intervals as start and end pairs in arbitrary order, merge every overlapping or touching pair and return the merged set sorted by start.

```
Input   [[1,3],[2,6],[8,10],[15,18]]     Output   [[1,6],[8,10],[15,18]]
Input   [[1,4],[4,5]]                    Output   [[1,5]]
```

Test Set H3

Senior Analyst band - medium to hard.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 7, b = 2, c = 5
c = a MOD b + b MOD c * a
Print c
```

A) 1 B) 3 C) 15 D) 8

Q2.

```
Integer f(Integer n)
  if(n <= 1)
    return 1
  End if
  if(n MOD 2 == 0)
    return f(n / 2)
  End if
  return f(n - 1) + 1
End function f()
Print f(20)
```

A) 3 B) 2 C) 4 D) 1

Q3.

```
Integer i, j, s
Set s = 0
for(each i from 1 to 4)
  Set j = i
  while(j > 0)
    s = s + (i * j)
    j = j - 2
  End while
End for
Print s
```

A) 30 B) 36 C) 45 D) 41

Q4.

```
Integer M[4][4], i, s
Set M = {{1,2,3,4},{5,6,7,8},{9,10,11,12},{13,14,15,16}}
Set s = 0
for(each i from 0 to 3)
  s = s + M[i][i] + M[i][3 - i]
End for
Print s
```

A) 34 B) 68 C) 64 D) 72

Q5.

```
Integer a, b, c, t, i
Set a = 1, b = 2, c = 3
for(each i from 1 to 4)
  t = a
  a = b + c
  c = b
  b = t
End for
Print a, b, c
```

A) 6 3 5 B) 8 3 6 C) 5 1 2 D) 8 6 3

Q6.

```
String s
Integer i, v, c
Set s = "SEQUENCE", v = 0, c = 0
for(each i from 0 to LENGTH(s) - 1)
  if(s[i] == 'A' OR s[i] == 'E' OR s[i] == 'I'
     OR s[i] == 'O' OR s[i] == 'U')
    v = v + 1
  else
    c = c + 1
  End if
End for
Print v - c
```

A) 2 B) 0 C) -2 D) 4

Q7.

```
Integer A[6], i, j, c
Set A = {5, 1, 4, 2, 8, 3}
Set c = 0
for(each i from 0 to 5)
  for(each j from i + 1 to 5)
    if(A[i] > A[j])
      c = c + 1
    End if
  End for
End for
Print c
```

A) 6 B) 8 C) 7 D) 9

Q8.

```
Integer n, s, d
Set n = 9875, s = 0
while(n > 0)
  d = n MOD 10
  n = n / 10
  if(d > 5)
    s = s * 10 + (d - 5)
  End if
End while
Print s
```

A) 234 B) 432 C) 23 D) 2340

Q9.

```
Integer f(Integer n, Integer a)
  if(n == 0)
    return a
  End if
  return f(n - 1, a * n)
End function f()
Print f(5, 1) / f(3, 1)
```

A) 24 B) 40 C) 6 D) 20

Q10.

```
Integer x, y
Set x = 22, y = 0
while(x != 0)
  y = y + (x AND 1)
  x = x >> 1
End while
Print y
```

A) 4 B) 3 C) 2 D) 5

Q11.

```
Integer A[5], i, p
Set A = {1, 2, 3, 4, 5}
Set p = 1
for(each i from 0 to 4)
  if(A[i] MOD 2 == 1)
    p = p * A[i]
  else
    p = p + A[i]
  End if
End for
Print p
```

A) 15 B) 120 C) 65 D) 45

Q12.

```
Integer i, j, c
Set c = 0, i = 0
while(i < 4)
  Set j = 0
  while(j < 4)
    if(i + j == 3 OR i == j)
      c = c + 1
    End if
    j = j + 1
  End while
  i = i + 1
End while
Print c
```

A) 7 B) 6 C) 4 D) 8

Q13.

```
Integer A[7], i, k, t, tmp
Set A = {1, 2, 3, 4, 5, 6, 7}
Set k = 3
for(each i from 0 to 6)
  t = (i + k) MOD 7
  if(t > i)
    tmp = A[i]
    A[i] = A[t]
    A[t] = tmp
  End if
End for
Print A[0], A[3], A[6]
```

A) 1 4 7 B) 4 7 1 C) 4 1 7 D) 5 2 3

Q14.

```
Integer n, r
Set n = 1, r = 0
repeat
  n = n * 2
  r = r + n
until(n >= 32)
Print n, r
```

A) 32 30 B) 64 126 C) 32 62 D) 16 30

Q15.

```
String s
Integer i, j, f
Set s = "ABCBA"
Set i = 0, j = LENGTH(s) - 1, f = 1
while(i < j)
  if(s[i] != s[j])
    f = 0
    break
  End if
  i = i + 1
  j = j - 1
End while
Print f, i
```

A) 1 0 B) 0 2 C) 1 5 D) 1 2

Q16.

```
Integer A[5], i, s, mx
Set A = {3, -2, 5, -1, 4}
Set s = 0, mx = 0
for(each i from 0 to 4)
  s = s + A[i]
  if(s < 0)
    s = 0
  End if
  if(s > mx)
    mx = s
  End if
End for
Print mx
```

A) 6 B) 9 C) 12 D) 5

Q17.

```
Integer f(Integer a, Integer b)
  if(a < b)
    return 0
  End if
  return 1 + f(a - b, b)
End function f()
Print f(47, 6)
```

A) 8 B) 6 C) 7 D) 5

Q18.

```
Integer A[4][3], i, j, mx, cs
Set A = {{3,8,1},{9,2,7},{4,6,5},{2,9,3}}
Set mx = 0
for(each j from 0 to 2)
  Set cs = 0
  for(each i from 0 to 3)
    cs = cs + A[i][j]
  End for
  if(cs > mx)
    mx = cs
  End if
End for
Print mx
```

A) 18 B) 16 C) 59 D) 25

Q19.

```
Integer n, c, i
Set n = 36, c = 0
for(each i from 1 to n)
  if(i * i <= n AND n MOD i == 0)
    c = c + 2
    if(i * i == n)
      c = c - 1
    End if
  End if
End for
Print c
```

A) 10 B) 9 C) 8 D) 6

Q20.

```
Integer A[6], i, j, t, n
Set A = {6, 5, 4, 3, 2, 1}
Set n = 6
for(each i from 1 to n - 1)
  t = A[i]
  Set j = i - 1
  while(j >= 0 AND A[j] > t)
    A[j + 1] = A[j]
    j = j - 1
  End while
  A[j + 1] = t
End for
Print A[0], A[5]
```

A) 6 1 B) 1 5 C) 2 6 D) 1 6

Section A - Computer Fundamentals (Q21 to Q30)

Q21. Appending to a dynamic array that doubles its capacity when full costs:

A) O(n) per append B) O(1) amortised C) O(log n) D) O(n log n)

Q22. Which statement about Java's finally block is FALSE?

A) It runs after try or catch completes B) It runs even if catch throws

C) It runs even after System.exit(0) D) It can override a return value

Q23. Per the C standard, sizeof(char) is:

A) Always 1 B) Platform-dependent C) 2 D) 4

Q24. R(A,B,C) with A as primary key and functional dependencies A to B and B to C is in:

A) 1NF only B) 2NF but not 3NF C) 3NF D) BCNF

Q25. Building a heap from an unsorted array using bottom-up heapify costs:

A) O(n log n) B) O(n) C) O(log n) D) O(n squared)

Q26. A page fault is classified as major when:

A) The page is resident but marked invalid

B) The page must be fetched from disk

C) The TLB missed D) The frame is dirty

Q27. A depth-first search on a directed graph reports a cycle upon encountering:

A) Any visited node B) A node currently on the recursion stack

C) A leaf D) A disconnected component

Q28. The static keyword on a file-scope C function means:

A) It cannot be called

B) Internal linkage, visible only in that translation unit

C) It is thread-safe D) Its locals persist

Q29. Which discipline prevents deadlock among threads acquiring multiple mutexes?

A) Randomised ordering B) A single consistent global acquisition order

C) Longer timeouts D) More mutexes

Q30. Which pair, taken together, uniquely reconstructs a binary tree?

A) Preorder plus postorder B) Inorder plus preorder

C) Level order alone D) Inorder alone

Section B - English Communication (Q31 to Q38)

Q31. The findings, ____ compelling, ____ replication before the team commits to a rewrite.

A) while / require B) while / requires

C) whilst / requiring D) although / requires

Q32. Synonym of EQUIVOCAL:

A) unambiguous B) ambiguous C) forceful D) verified

Q33. Antonym of SALIENT:

A) prominent B) inconspicuous C) striking D) notable

Q34. "To move the goalposts" means:

A) Change the criteria after work has begun B) Relocate a resource

C) Increase the budget D) Reassign ownership

Q35. Find the error: "The data suggests that latency spikes correlates with cache evictions."

A) The data suggests B) that latency spikes C) correlates with D) No error

Q36. Which uses the subjunctive correctly?

A) If the index was present, the query would be fast

B) If the index were present, the query would be fast

C) If the index is present, the query would be fast

D) If the index were present, the query will be fast

Q37. Rearrange: (P) that the fix is safe (Q) a passing test suite (R) is not evidence (S) only that the tested paths behave

A) QRPS B) PQRS C) QPRS D) RQPS

Q38. Read the passage and answer.

```
The observability argument is often framed as logs versus metrics versus
traces, as though a team must choose. In practice the three answer different
questions: metrics tell you that something changed, traces tell you where,
and logs tell you what. A team that invests in only one will find itself able
to detect an incident it cannot explain, or explain one it detected too late.
```

The passage concludes the three are:

A) Interchangeable B) Complementary rather than competing

C) Equally expensive D) Redundant at scale

Section C - Game-Based Cognitive (Q39 to Q44)

Q39. switchChallenge, single operator.

```
Input     SQR   STR   DIA   CIR
Output    DIA   CIR   SQR   STR
```

A) 4312 B) 3421 C) 3412 D) 1342

Q40. switchChallenge, two layers. Find Operator 2.

```
Input         SQR   STR   DIA   CIR
Operator 1    4231
Operator 2    ?
Output        STR   DIA   SQR   CIR
```

A) 3412 B) 2341 C) 4123 D) 1432

Q41. switchChallenge, three layers. Find the middle operator.

```
Input         SQR   STR   DIA   CIR
Operator 1    1324
Operator 2    ?
Operator 3    2413
Output        SQR   STR   CIR   DIA
```

A) 2314 B) 3241 C) 1432 D) 4123

Q42. digitChallenge. Digits 1 to 9, each used only once in the equation.

```
_  /  _  +  _  *  _  =  29
```

A) 9, 3, 8, 2 B) 6, 3, 9, 3 C) 8, 4, 9, 3 D) 8, 2, 5, 6

Q43. gridChallenge. Work the three stages in order.

Stage 1 - memorise this grid for 10 seconds, then cover it.

```
X   .   .   .
.   .   X   .
.   X   .   .
.   .   .   X
```

Stage 2 - distractor. Grid A rotated 180 degrees produces which option?

```
Grid A            Option A          Option B
X   X   .         .   .   .         .   X   X
.   .   X         X   .   .         X   .   .
.   .   .         .   X   X         .   .   .

                  Option C          Option D
                  .   .   .         X   X   .
                  .   .   X         X   .   .
                  X   X   .         .   .   .
```

A) Option A B) Option B C) Option C D) Option D

Stage 3 - recall. In the Stage 1 grid, which row held a dot in column 2?

Q44. motionChallenge, discrete tick. A ball at row 3, column 1 of a 5 by 5 grid moves diagonally up and to the right, reflecting off walls. Position after 6 ticks?

A) row 4, col 4 B) row 5, col 3 C) row 1, col 3 D) row 3, col 5

Section D - Coding Round (45 minutes)

Problem 1 - Longest Consecutive Sequence

Given an unsorted integer array, return the length of the longest run of consecutive integers present. Order in the array is irrelevant. O(n) expected; sorting is O(n log n) and loses marks.

```
Input   [100, 4, 200, 1, 3, 2]          Output   4    (1, 2, 3, 4)
Input   [9, 1, 4, 7, 3, 2, 6, 8, 5]     Output   9
```

Problem 2 - Kth Largest Element

Given an unsorted array and k, return the k-th largest element WITHOUT fully sorting the array. Target O(n log k) with a min-heap of size k, or average O(n) with quickselect.

```
Input   [3, 2, 1, 5, 6, 4], k = 2               Output   5
Input   [3, 2, 3, 1, 2, 4, 5, 5, 6], k = 4      Output   4
```

Test Set H4

Senior Analyst band - medium to hard.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 9, b = 5, c = 2
b = a / c * c
c = a - b
Print b, c
```

A) 9 0 B) 8 1 C) 10 -1 D) 8 4

Q2.

```
Integer f(Integer n)
  if(n == 1)
    return 0
  End if
  return 1 + f(n / 2)
End function f()
Print f(100)
```

A) 5 B) 7 C) 6 D) 8

Q3.

```
Integer i, j, s
Set s = 0, i = 2
while(i <= 8)
  Set j = i
  while(j <= 8)
    s = s + 1
    j = j + i
  End while
  i = i + 2
End while
Print s
```

A) 7 B) 8 C) 9 D) 10

Q4.

```
Integer M[3][3], i, j, c
Set M = {{1,0,1},{0,1,0},{1,1,0}}
Set c = 0
for(each i from 0 to 2)
  for(each j from 0 to 2)
    if(M[i][j] == 1 AND M[j][i] == 1)
      c = c + 1
    End if
  End for
End for
Print c
```

A) 3 B) 5 C) 4 D) 6

Q5.

```
Integer a, b, c, i
Set a = 2, b = 1, c = 0
for(each i from 1 to 5)
  c = a * b - c
  a = b
  b = c
End for
Print a, b, c
```

A) 0 0 0 B) 2 1 0 C) 1 2 2 D) 2 0 0

Q6.

```
String s
Integer i, c
Set s = "PARALLEL", c = 0
for(each i from 0 to LENGTH(s) - 1)
  if(s[i] == 'L' OR s[i] == 'A')
    c = c + 1
  End if
End for
Print c
```

A) 4 B) 5 C) 6 D) 3

Q7.

```
Integer A[6], i, j, t
Set A = {1, 5, 3, 2, 4, 6}
for(each i from 0 to 5)
  for(each j from 0 to 4)
    if(A[j] > A[j+1])
      t = A[j]
      A[j] = A[j+1]
      A[j+1] = t
    End if
  End for
End for
Print A[1], A[4]
```

A) 5 4 B) 2 5 C) 1 6 D) 3 4

Q8.

```
Integer n, s, d, c
Set n = 4823, s = 0, c = 0
while(n > 0)
  d = n MOD 10
  n = n / 10
  c = c + 1
  if(c MOD 2 == 1)
    s = s + d
  else
    s = s - d
  End if
End while
Print s
```

A) 3 B) 5 C) 9 D) 17

Q9.

```
Integer f(Integer n)
  if(n <= 0)
    return 1
  End if
  return f(n - 1) * 2 - 1
End function f()
Print f(4)
```

A) 16 B) 15 C) 1 D) 7

Q10.

```
Integer a, b, c
Set a = 6, b = 10
c = a XOR b
a = a AND b
b = a OR c
Print a, b, c
```

A) 2 14 12 B) 6 10 12 C) 2 12 14 D) 14 2 12

Q11.

```
Integer A[5], i, s
Set A = {2, 4, 6, 8, 10}
Set s = 0
for(each i from 0 to 4)
  if(i > 0 AND A[i] > A[i-1])
    s = s + A[i] - A[i-1]
  End if
End for
Print s
```

A) 30 B) 8 C) 10 D) 2

Q12.

```
Integer i, j, c
Set c = 0
for(each i from 1 to 5)
  Set j = 1
  while(j <= i)
    if((i + j) MOD 2 == 0)
      c = c + 1
    End if
    j = j + 1
  End while
End for
Print c
```

A) 8 B) 9 C) 10 D) 15

Q13.

```
Integer A[7], i, lo, hi, mid, key
Set A = {2, 4, 6, 8, 10, 12, 14}
Set lo = 0, hi = 6, key = 9, i = 0
while(lo <= hi)
  mid = (lo + hi) / 2
  i = i + 1
  if(A[mid] == key)
    break
  End if
  if(A[mid] < key)
    lo = mid + 1
  else
    hi = mid - 1
  End if
End while
Print i, lo
```

A) 3 4 B) 4 4 C) 3 5 D) 2 4

Q14.

```
Integer n, r, c
Set n = 5, r = 1, c = 0
while(c < 4)
  r = r * n - c
  c = c + 1
End while
Print r
```

A) 625 B) 587 C) 590 D) 118

Q15.

```
String s
Integer i, mx, cur
Set s = "AABBBBCC"
Set mx = 1, cur = 1
for(each i from 1 to LENGTH(s) - 1)
  if(s[i] == s[i-1])
    cur = cur + 1
  else
    cur = 1
  End if
  if(cur > mx)
    mx = cur
  End if
End for
Print mx
```

A) 2 B) 3 C) 4 D) 8

Q16.

```
Integer A[5], i, j, c, f
Set A = {3, 1, 4, 1, 5}
Set c = 0
for(each i from 0 to 4)
  Set j = 0, f = 0
  while(j < i)
    if(A[j] == A[i])
      f = 1
    End if
    j = j + 1
  End while
  if(f == 0)
    c = c + 1
  End if
End for
Print c
```

A) 5 B) 3 C) 4 D) 2

Q17.

```
Integer f(Integer n)
  if(n < 3)
    return n
  End if
  return f(n - 1) - f(n - 2) + f(n - 3)
End function f()
Print f(6)
```

A) 1 B) 0 C) 2 D) 3

Q18.

```
Integer M[4][4], i, j, t
Set M = {{1,2,3,4},{5,6,7,8},{9,10,11,12},{13,14,15,16}}
for(each i from 0 to 1)
  for(each j from 0 to 3)
    t = M[i][j]
    M[i][j] = M[3 - i][j]
    M[3 - i][j] = t
  End for
End for
Print M[0][0], M[1][1], M[3][3]
```

A) 13 10 4 B) 1 6 16 C) 13 6 4 D) 4 10 13

Q19.

```
Integer n, m, g, l, t
Set n = 12, m = 18
Set g = n, l = m
while(l != 0)
  t = l
  l = g MOD l
  g = t
End while
Print g, (n * m) / g
```

A) 6 36 B) 6 216 C) 2 108 D) 18 12

Q20.

```
Integer A[8], lo, hi, mid
Set A = {5, 6, 7, 1, 2, 3, 4, 5}
Set lo = 0, hi = 7
while(lo < hi)
  mid = (lo + hi) / 2
  if(A[mid] > A[hi])
    lo = mid + 1
  else
    hi = mid
  End if
End while
Print lo, A[lo]
```

A) 3 1 B) 0 5 C) 4 2 D) 3 7

Section A - Computer Fundamentals (Q21 to Q30)

Q21. Which sort performs best on a nearly sorted array?

A) QuickSort with first-element pivot B) Insertion sort

C) Heap sort D) Selection sort

Q22. In Java, the == operator applied to two Integer objects each holding 1000 compares:

A) Values B) References C) Hash codes D) It is always true

Q23. Which C++ smart pointer provides shared ownership via reference counting?

A) unique_ptr B) shared_ptr C) weak_ptr D) auto_ptr

Q24. A composite index on (A, B) efficiently serves a query filtering only on:

A) B alone B) A alone C) Neither D) Both individually

Q25. Worst-case auxiliary space of DFS on a graph with V vertices:

A) O(1) B) O(V) C) O(V squared) D) O(log V)

Q26. With all burst times known in advance, which scheduler minimises average waiting time?

A) FCFS B) Round Robin C) Shortest Job First D) Priority

Q27. An IPv4 /26 subnet provides how many usable host addresses?

A) 64 B) 62 C) 30 D) 126

Q28. Given only a pointer to a node, which operation is O(1) on a doubly linked list but O(n) on a singly linked list?

A) Insert after it B) Traverse C) Delete that node D) Search

Q29. In C, char *s = "abc"; followed by s[0] = 'x'; is:

A) Valid B) Undefined behaviour - string literals are not writable

C) Always a compile error D) Valid but slow

Q30. Which protocol guarantees serialisable schedules?

A) Timestamp ordering alone B) Two-phase locking

C) Read Committed D) Optimistic reads

Section B - English Communication (Q31 to Q38)

Q31. Each of the microservices, along with its sidecar proxy, ____ independently deployable.

A) are B) is C) have been D) were

Q32. Synonym of SPURIOUS:

A) genuine B) counterfeit C) urgent D) abundant

Q33. Antonym of EXACERBATE:

A) aggravate B) alleviate C) intensify D) provoke

Q34. "To bikeshed" means:

A) Argue over trivia while ignoring the substantive issue

B) Build a prototype C) Store unused code D) Delay indefinitely

Q35. Find the error: "Between you and I, the estimate was never realistic."

A) Between you and I B) the estimate C) was never realistic D) No error

Q36. Which sentence uses "comprise" correctly?

A) The system is comprised of four services

B) The system comprises four services

C) The system comprises of four services

D) Four services comprise into the system

Q37. Rearrange: (P) the bug was reproducible (Q) it took three weeks (R) but only under load (S) to construct a load harness

A) PRQS B) QSPR C) PQRS D) RPQS

Q38. Read the passage and answer.

```
Refactoring is often justified by future velocity, a benefit that is real but
unfalsifiable in any single case. The team that refactors and then ships
faster cannot prove the refactor caused it; the team that skips it and slows
down cannot prove the omission was the reason. The argument for refactoring
therefore rests on accumulated experience rather than local measurement.
```

The justification for refactoring is characterised as:

A) Unfounded

B) Supported inductively rather than demonstrable case by case

C) Purely aesthetic D) Measurable with better tooling

Section C - Game-Based Cognitive (Q39 to Q44)

Q39. switchChallenge, single operator.

```
Input     TRI   SQR   CIR   STR
Output    STR   CIR   TRI   SQR
```

A) 3421 B) 4321 C) 2413 D) 4312

Q40. switchChallenge, two layers, with the FIRST operator missing. Operator 2 is given; deduce Operator 1 by first inverting Operator 2 to recover the hidden intermediate row.

```
Input         TRI   SQR   CIR   STR
Operator 1    ?
Operator 2    2143
Output        STR   CIR   TRI   SQR
```

A) 4312 B) 3421 C) 2341 D) 1432

Q41. switchChallenge, three layers. Find the middle operator.

```
Input         TRI   SQR   CIR   STR
Operator 1    2314
Operator 2    ?
Operator 3    4123
Output        CIR   TRI   SQR   STR
```

A) 4213 B) 2431 C) 3142 D) 1324

Q42. digitChallenge. Digits 1 to 9, each used only once in the equation.

```
_  *  _  +  _  -  _  =  30
```

A) 6, 5, 3, 3 B) 8, 3, 9, 2 C) 7, 4, 5, 3 D) 9, 2, 7, 1

Q43. gridChallenge. Work the three stages in order.

Stage 1 - memorise this grid for 10 seconds, then cover it.

```
.   .   .   X
X   .   .   .
.   X   X   .
.   .   .   .
```

Stage 2 - distractor. Is this grid symmetric about the main diagonal, running from top-left to bottom-right?

```
X   .   X
.   X   .
X   .   X
```

Stage 3 - recall. How many dots were in row 3 of the Stage 1 grid?

Q44. motionChallenge, discrete tick. A ball at row 1, column 4 of a 4 by 4 grid moves diagonally down and to the left, reflecting off walls. Position after 5 ticks?

A) row 4, col 1 B) row 3, col 2 C) row 2, col 3 D) row 1, col 4

Section D - Coding Round (45 minutes)

Problem 1 - Product of Array Except Self

For each index i, output the product of every element except the one at i. Division is NOT allowed. O(n) time; the output array does not count toward extra space.

```
Input   [1, 2, 3, 4]             Output   [24, 12, 8, 6]
Input   [-1, 1, 0, -3, 3]        Output   [0, 0, 9, 0, 0]
```

Problem 2 - Set Matrix Zeroes

Given an m by n matrix, if any element is 0, set that element's entire row and column to 0. Do it in place with O(1) extra space. Zeroing as you scan is wrong because it cascades.

```
Input       Output
1 1 1       1 0 1
1 0 1       0 0 0
1 1 1       1 0 1
```

Test Set H5

Senior Analyst band - medium to hard.

Section A - Pseudo-code (Q1 to Q20)

Q1.

```
Integer a, b, c
Set a = 17, b = 6, c = 4
c = a MOD b MOD c
b = a / b / c
Print b, c
```

A) 2 1 B) 1 2 C) 2 5 D) 4 1

Q2.

```
Integer f(Integer n)
  if(n == 0)
    return 0
  End if
  if(n MOD 3 == 0)
    return f(n - 1) + 3
  End if
  return f(n - 1) + 1
End function f()
Print f(9)
```

A) 9 B) 12 C) 15 D) 18

Q3.

```
Integer i, j, c
Set c = 0, i = 1
while(i <= 16)
  Set j = i
  while(j <= 16)
    c = c + 1
    j = j * 2
  End while
  i = i * 4
End while
Print c
```

A) 8 B) 9 C) 10 D) 12

Q4.

```
Integer M[3][3], i, j, s, mn
Set M = {{4,1,7},{2,8,3},{6,5,9}}
Set s = 0
for(each i from 0 to 2)
  Set mn = M[i][0]
  for(each j from 1 to 2)
    if(M[i][j] < mn)
      mn = M[i][j]
    End if
  End for
  s = s + mn
End for
Print s
```

A) 8 B) 12 C) 6 D) 24

Q5.

```
Integer p, q, r, i
Set p = 1, q = 0, r = 0
for(each i from 1 to 5)
  r = p + q
  q = p
  p = r
End for
Print p, q, r
```

A) 8 5 8 B) 5 3 5 C) 8 5 13 D) 13 8 13

Q6.

```
String s
Integer i, c
Set s = "ABABABA", c = 0
for(each i from 0 to LENGTH(s) - 3)
  if(s[i] == s[i+2])
    c = c + 1
  End if
End for
Print c
```

A) 3 B) 4 C) 5 D) 7

Q7.

```
Integer A[6], i, e, o
Set A = {7, 2, 9, 4, 1, 6}
Set e = 0, o = 0
for(each i from 0 to 5)
  if(A[i] MOD 2 == 0)
    e = e + A[i]
  else
    o = o * 10 + A[i]
  End if
End for
Print e, o
```

A) 12 791 B) 12 17 C) 6 791 D) 12 197

Q8.

```
Integer n, c
Set n = 1, c = 0
while(c < 5)
  n = n * 3 + c
  c = c + 1
End while
Print n
```

A) 243 B) 301 C) 297 D) 99

Q9.

```
Integer f(Integer n, Integer k)
  if(k == 0 OR k == n)
    return 1
  End if
  return f(n - 1, k - 1) + f(n - 1, k)
End function f()
Print f(5, 2)
```

A) 5 B) 10 C) 20 D) 15

Q10.

```
Integer a, b
Set a = 20, b = 12
Print (a AND b), (a OR b), (a XOR b)
```

A) 4 28 24 B) 12 20 8 C) 4 24 28 D) 8 28 20

Q11.

```
Integer A[5], i, j
Set A = {10, 20, 30, 40, 50}
Set i = 0
while(i < 5)
  Set j = i + 1
  while(j < 5)
    if(A[i] + A[j] == 60)
      Print i, j
    End if
    j = j + 1
  End while
  i = i + 1
End while
```

A) 0 4 only B) 0 4 and 1 3 C) 1 3 only D) No output

Q12.

```
Integer n, s, t
Set n = 7, s = 0
Set t = n
while(t > 0)
  s = s + t * t
  t = t - 2
End while
Print s
```

A) 84 B) 140 C) 49 D) 91

Q13.

```
String s1, s2
Integer i, c1, c2
Set s1 = "SILENT", s2 = "LISTEN"
Set c1 = 0, c2 = 0
for(each i from 0 to 5)
  c1 = c1 + (s1[i] - 'A')
  c2 = c2 + (s2[i] - 'A')
End for
if(c1 == c2)
  Print "MATCH"
else
  Print c1 - c2
End if
```

A) MATCH B) 0 C) 6 D) -6

Q14.

```
Integer A[6], i, k, c
Set A = {1, 2, 3, 4, 5, 6}
Set k = 2, c = 0
for(each i from 0 to 5)
  if(A[i] MOD k == 0)
    c = c + A[i]
    k = k + 1
  End if
End for
Print c, k
```

A) 20 7 B) 12 4 C) 21 7 D) 20 6

Q15.

```
Integer M[3][3], i, j, s1, s2
Set M = {{2,7,6},{9,5,1},{4,3,8}}
Set s2 = 0
for(each i from 0 to 2)
  Set s1 = 0
  for(each j from 0 to 2)
    s1 = s1 + M[i][j]
  End for
  if(s1 != 15)
    s2 = s2 + 1
  End if
End for
Print s2
```

A) 0 B) 1 C) 3 D) 15

Q16.

```
Integer f(Integer n)
  if(n <= 1)
    return n
  End if
  Integer a, b, c, i
  Set a = 0, b = 1
  for(each i from 2 to n)
    c = a + b
    a = b
    b = c
  End for
  return b
End function f()
Print f(8)
```

A) 13 B) 21 C) 34 D) 8

Q17.

```
Integer A[7], i, w
Set A = {1, 1, 2, 2, 2, 3, 3}
Set w = 0
for(each i from 0 to 6)
  if(i == 0 OR A[i] != A[i-1])
    A[w] = A[i]
    w = w + 1
  End if
End for
Print w, A[0], A[1], A[2]
```

A) 3 1 2 3 B) 7 1 1 2 C) 3 1 2 2 D) 4 1 2 3

Q18.

```
Integer n, r, t
Set n = 1234, r = 0
Set t = n
while(t > 0)
  r = r * 10 + (t MOD 10)
  t = t / 10
End while
if(r == n)
  Print "PALINDROME"
else
  Print r - n
End if
```

A) PALINDROME B) 3087 C) 4321 D) 1234

Q19.

```
Integer A[8], lo, hi, mid, key, r
Set A = {1, 3, 5, 7, 9, 11, 13, 15}
Set lo = 0, hi = 7, key = 11, r = 0
while(lo <= hi)
  mid = lo + (hi - lo) / 2
  r = r + 1
  if(A[mid] == key)
    break
  End if
  if(A[mid] < key)
    lo = mid + 1
  else
    hi = mid - 1
  End if
End while
Print r, mid
```

A) 2 5 B) 3 5 C) 2 4 D) 1 3

Q20.

```
Integer A[5], B[5], i, j
Set A = {1, 2, 3, 4, 5}
Set j = 0
for(each i from 4 down to 0)
  if(A[i] MOD 2 == 1)
    B[j] = A[i]
    j = j + 1
  End if
End for
for(each i from 4 down to 0)
  if(A[i] MOD 2 == 0)
    B[j] = A[i]
    j = j + 1
  End if
End for
Print B[0], B[2], B[4]
```

A) 5 1 2 B) 1 3 5 C) 5 3 1 D) 1 5 2

Section A - Computer Fundamentals (Q21 to Q30)

Q21. Lower bound on comparisons for any comparison-based sort of n elements:

A) Omega(n) B) Omega(n log n) C) Omega(log n) D) Omega(n squared)

Q22. An unchecked exception propagating out of Java's main results in:

A) A compile error

B) The JVM printing the stack trace and terminating the thread

C) Silent suppression D) main restarting

Q23. In C++, const placed after a member function's parameter list means:

A) The return value is const

B) The function cannot modify non-mutable members

C) The function is static D) Parameters are const

Q24. Which anomaly occurs under Read Committed but not Repeatable Read?

A) Dirty read B) Non-repeatable read C) Phantom read D) Lost update

Q25. Extracting the root from a min-heap and restoring the heap property costs:

A) O(1) B) O(log n) C) O(n) D) O(n log n)

Q26. Copy-on-write in fork() means:

A) Pages are duplicated immediately

B) Pages stay shared until one process writes

C) The child cannot write D) Memory is swapped to disk

Q27. Which IPv4 header field stops packets circulating indefinitely?

A) Checksum B) TTL C) Flags D) Protocol

Q28. Topological sort is defined for:

A) Any graph B) Directed acyclic graphs

C) Undirected trees D) Complete graphs

Q29. For int a[5]; the expression sizeof(a) / sizeof(a[0]) evaluates to:

A) 5 B) 20 C) 4 D) 1

Q30. Which normal form's decomposition is not always dependency-preserving?

A) 1NF B) 2NF C) 3NF D) BCNF

Section B - English Communication (Q31 to Q38)

Q31. The set of assumptions on which the design rests ____ never been documented, which ____ why reviewers disagree.

A) have / explain B) has / explains

C) have / explains D) has / explain

Q32. Synonym of INCHOATE:

A) complete B) rudimentary C) refined D) urgent

Q33. Antonym of CIRCUMSPECT:

A) cautious B) reckless C) prudent D) wary

Q34. "To gold-plate" a feature means:

A) Add refinement beyond what was requested B) Make it expensive

C) Document it heavily D) Deprecate it

Q35. Find the error: "Neither of the two candidates were able to explain their own design."

A) Neither of B) the two candidates C) were able D) No error

Q36. Which sentence is free of a misplaced modifier?

A) We only deployed the fix to staging, not production

B) We deployed the fix only to staging, not production

C) Only we deployed the fix to staging, not production

D) We deployed only the fix to staging, not production

Q37. Rearrange: (P) the incident review concluded (Q) not the deploy that exposed it (R) that the root cause (S) was the missing constraint

A) PRSQ B) PQRS C) RSPQ D) PRQS

Q38. Read the passage and answer.

```
A postmortem that names an individual has already failed. The question is not
who typed the command but why the system permitted a single command to cause
that much damage. Blame terminates inquiry at the most convenient point, which
is almost never the most informative one.
```

The author objects to naming individuals chiefly because it:

A) Is unkind B) Halts investigation before the systemic cause surfaces

C) Carries legal risk D) Degrades documentation

Section C - Game-Based Cognitive (Q39 to Q44)

Q39. switchChallenge, single operator.

```
Input     CIR   STR   SQR   TRI
Output    SQR   TRI   CIR   STR
```

A) 4312 B) 3142 C) 3412 D) 2341

Q40. switchChallenge, two layers. Find Operator 2.

```
Input         CIR   STR   SQR   TRI
Operator 1    4213
Operator 2    ?
Output        CIR   TRI   SQR   STR
```

A) 2413 B) 3142 C) 4321 D) 1324

Q41. switchChallenge, three layers. Find the middle operator. The output equals the input, so the three operators compose to the identity.

```
Input         CIR   STR   SQR   TRI
Operator 1    4321
Operator 2    ?
Operator 3    3412
Output        CIR   STR   SQR   TRI
```

A) 4132 B) 2143 C) 1324 D) 3412

Q42. digitChallenge. Digits 1 to 9, each used only once in the equation.

```
_  /  _  *  _  -  _  =  19
```

A) 8, 2, 6, 5 B) 6, 2, 7, 2 C) 9, 3, 8, 4 D) 6, 3, 9, 2

Q43. gridChallenge. Work the three stages in order.

Stage 1 - memorise this grid for 12 seconds, then cover it.

```
X   .   .   X
.   .   X   .
.   X   .   .
X   .   .   .
```

Stage 2 - distractor. Grid A rotated 90 degrees clockwise produces which option?

```
Grid A           Option A         Option B
X   .   .        .   X   X        X   X   .
X   X   .        .   X   .        .   X   .
.   .   X        X   .   .        .   .   X

                 Option C         Option D
                 X   .   .        .   .   X
                 .   X   X        .   X   .
                 .   .   X        X   X   .
```

A) Option A B) Option B C) Option C D) Option D

Stage 3 - recall. How many dots were in column 1 of the Stage 1 grid?

Q44. motionChallenge, discrete tick. A ball at row 3, column 3 of a 5 by 5 grid moves diagonally up and to the left, reflecting off walls. Position after 7 ticks?

A) row 5, col 5 B) row 4, col 4 C) row 3, col 3 D) row 1, col 1

Section D - Coding Round (45 minutes)

Problem 1 - Search in Rotated Sorted Array

A sorted array of distinct integers has been rotated an unknown number of times. Given a target, return its index or -1. Must run in O(log n); a linear scan fails the constraint.

```
Input   [4, 5, 6, 7, 0, 1, 2], target = 0     Output   4
Input   [4, 5, 6, 7, 0, 1, 2], target = 3     Output   -1
```

Problem 2 - Decode Ways

A message of digits was encoded with A equal to 1 through Z equal to 26. Return the number of distinct ways to decode it. Leading zeros make a grouping invalid, so "06" cannot be read as 6. Dynamic programming, O(n) time.

```
Input   "12"      Output   2     (AB, L)
Input   "226"     Output   3     (BZ, VF, BBF)
Input   "06"      Output   0
```

Answer Keys

Section A runs Q1 to Q30, Section B runs Q31 to Q40 in Sets 1 to 5 and Q31 to Q38 in Sets H1 to H5, and Section C occupies the remaining numbers. Section D is open-ended and is graded on test cases passed and time complexity, so no key is given.

Every pseudo-code answer in this book was traced from first principles rather than copied from a published key.

Set 1

```
Section A 1-B 2-C 3-A 4-B 5-B 6-A 7-A 8-B 9-B 10-B
11-A 12-B 13-A 14-B 15-A 16-B 17-A 18-B 19-B 20-B
21-B 22-C 23-C 24-B 25-B 26-C 27-C 28-B 29-C 30-B
Section B 31-C 32-B 33-C 34-B 35-B 36-A 37-A 38-A 39-B 40-C
Section C 41-A (CIR) 42-B 5 * 3 - 2 = 13 43-A (6 steps) 44-B (42)
```

Set 2

```
Section A 1-B 2-B 3-B 4-A 5-B 6-B 7-B 8-A 9-A 10-B
11-A 12-A 13-B 14-A 15-A 16-B 17-A 18-A 19-B 20-A
21-B 22-B 23-C 24-B 25-A 26-C 27-C 28-B 29-B 30-B
Section B 31-B 32-B 33-B 34-B 35-B 36-B 37-A 38-A 39-B 40-C
Section C 41-D (PLS) 42-A 8 / 4 + 3 = 5 43-C (48) 44-B (5)
```

Set 3

```
Section A 1-A 2-B 3-A 4-C 5-B 6-B 7-B 8-B 9-B 10-B
11-B 12-A 13-B 14-A 15-A 16-B 17-B 18-B 19-B 20-B
21-C 22-B 23-C 24-C 25-B 26-A 27-B 28-B 29-B 30-B
Section B 31-C 32-B 33-C 34-B 35-B 36-A 37-B 38-A 39-B 40-B
Section C 41-B (TRI) 42-B 6 * 2 - 7 = 5 43-D (95) 44-C (51)
```

Set 4

```
Section A 1-A 2-B 3-B 4-B 5-B 6-A 7-B 8-B 9-A 10-A
11-A 12-A 13-C 14-A 15-B 16-A 17-A 18-A 19-B 20-A
21-B 22-B 23-C 24-B 25-B 26-B 27-B 28-C 29-B 30-B
Section B 31-B 32-B 33-B 34-B 35-C 36-A 37-A 38-A 39-B 40-B
Section C 41-C (SQR) 42-A 3 + 4 * 6 = 27 43-C (37) 44-A (8 steps)
```

Set 5

```
Section A 1-B 2-A 3-B 4-A 5-B 6-B 7-C 8-B 9-B 10-A
11-A 12-B 13-B 14-B 15-A 16-A 17-B 18-A 19-B 20-B
21-A 22-B 23-B 24-B 25-B 26-B 27-B 28-B 29-B 30-B
Section B 31-B 32-B 33-B 34-B 35-C 36-A 37-A 38-A 39-A 40-B
Section C 41-A (CIR) 42-A 20 / 5 - 3 = 1 43-C (720) 44-C (22)
```

Set H1

```
Section A 1-C 2-A 3-B 4-D 5-B 6-C 7-B 8-A 9-C 10-B
11-D 12-A 13-B 14-C 15-B 16-D 17-A 18-B 19-C 20-B
21-B 22-B 23-C 24-C 25-C 26-B 27-B 28-C 29-B 30-B
Section B 31-C 32-B 33-B 34-A 35-C 36-B 37-A 38-B
Section C 39-B 40-C 41-A 42-A 44-A (row 3, col 3)
43 Stage 2 - Yes, they are mirror images. Stage 3 - 2 dots.
```

Set H2

Section A 1-C 2-D 3-B 4-C 5-D 6-B 7-C 8-C 9-B 10-D

11-B 12-D 13-C 14-D 15-D 16-B 17-A 18-C 19-B 20-D

21-B 22-B 23-B 24-B 25-C 26-C 27-D 28-C 29-B 30-C

Section B 31-B 32-B 33-B 34-A 35-C 36-B 37-A 38-B

Section C 39-C 40-D 41-B 42-A 44-A (row 1, col 2)

43 Stage 2 - Yes, they are mirror images. Stage 3 - 2 dots.

Set H3

Section A 1-C 2-B 3-D 4-B 5-D 6-B 7-C 8-A 9-D 10-B

11-C 12-D 13-B 14-C 15-D 16-B 17-C 18-D 19-B 20-D

21-B 22-C 23-A 24-B 25-B 26-B 27-B 28-B 29-B 30-B

Section B 31-A 32-B 33-B 34-A 35-C 36-B 37-A 38-B

Section C 39-C 40-B 41-D 42-C 44-B (row 5, col 3)

43 Stage 2 - Option A. Stage 3 - row 3.

Set H4

Section A 1-B 2-C 3-B 4-C 5-A 6-B 7-B 8-B 9-C 10-A

11-B 12-B 13-A 14-B 15-C 16-C 17-C 18-A 19-A 20-A

21-B 22-B 23-B 24-B 25-B 26-C 27-B 28-C 29-B 30-B

Section B 31-B 32-B 33-B 34-A 35-A 36-B 37-A 38-B

Section C 39-D 40-B 41-C 42-C 44-C (row 2, col 3)

43 Stage 2 - Yes, symmetric about the main diagonal. Stage 3 - 2 dots.

Set H5

Section A 1-A 2-C 3-B 4-A 5-A 6-C 7-A 8-B 9-B 10-A

11-B 12-A 13-A 14-A 15-A 16-B 17-A 18-B 19-A 20-A

21-B 22-B 23-B 24-B 25-B 26-B 27-B 28-B 29-A 30-D

Section B 31-B 32-B 33-B 34-A 35-C 36-B 37-A 38-B

Section C 39-C 40-B 41-B 42-A 44-B (row 4, col 4)

43 Stage 2 - Option A. Stage 3 - 2 dots.

Traps Worth Reviewing

If you missed any of the following, the error is conceptual rather than careless, and it is worth re-tracing the item on paper before moving on.

Pass-by-value

Set H1 Q6 and Set H2 Q5. A function that assigns to its parameter changes nothing in the caller. In H2 Q5 the function is even named swap, which is the whole trap; x and y come back unchanged and only the returned sum reflects the internal work.

Fixed points and collapses

Set H4 Q9. The recurrence doubles then subtracts one, and f(1) evaluates to 1, so every later call reproduces 1. The answer is 1, not 15 or 16. Set H4 Q5 collapses to all zeros by the third iteration. Set H1 Q13 reverses the array twice, once on the first two iterations and again on the last two, returning the original order.

Integer division

Set 1 Q7, Set H1 Q1, Set H5 Q1. Division discards the fractional part before the next operation consumes it. In H1 Q1, c divided by b is 1 and not 1.33, which changes the entire result.

Loop exit conditions

Set 1 Q16, Set H2 Q3, Set H2 Q14. Count iterations by identifying when the condition first fails, not when it nearly fails. In H2 Q14 the break fires after the addition, so the loop variable has already advanced past the value you might expect.

Bit patterns versus values

Set H2 Q10 reverses the bit pattern of 29, giving 10111 which is 23, rather than preserving the value. Set H1 Q10 rebuilds 45 unchanged while the counter records the bit width, so the two printed values measure different things.

Binary search variants

Set H2 Q17 is a lower-bound search and returns the FIRST index holding the key, not any index. Set H4 Q20 finds the minimum of a rotated array and compares against A[hi], not A[lo]. Set 5 Q6 counts iterations for a key that is absent, so the loop runs to exhaustion.

Counting problems

Set H3 Q7 counts inversions, not comparisons. Set H3 Q19 counts the divisors of 36, which is 9, by pairing each divisor below the square root with its partner and applying the correction when i squared equals n.

Structural shortcuts

Set H5 Q13 relies on SILENT and LISTEN being anagrams, so the character-code sums must be equal and you never need to compute either one. Set H5 Q15 is a magic square, every row sums to 15, so the counter never increments. Set H5 Q11 prints twice, not once.

switchChallenge inversion

Set H2 Q40, Set H4 Q40 and Set H5 Q41 all reduce to inverting an operator rather than tracing every layer. Set H2 Q40 asks for the inverse of 3142, which is 2413. Set H4 Q40 hides the FIRST operator, so invert the given second operator to recover the intermediate row, then read off the first. Set H5 Q41 composes to the identity.

digitChallenge distinctness

In every digitChallenge item across all ten sets, at least one wrong option reaches the target value while reusing a digit. Set H1 Q42 option B, Set H2 Q42 option B, Set H3 Q42 option B, Set H4 Q42 option A and Set H5 Q42 options B and D are all of this kind. Screen for distinctness before you compute.

Sources And Attribution

The assessment pattern, section structure, timings, topic coverage and band mapping in this book were compiled from the following public sources. All questions themselves are original and were written for this book; no question text was copied from any source.

Pattern, syllabus and process

Capgemini official Exceller page for technical graduates, capgemini.com, careers section for students and graduates.

PrepInsta - Capgemini Exceller syllabus and test pattern, recruitment process, coding questions, and game-based aptitude pages.

PlacementPreparation - Capgemini latest syllabus and test pattern, and recruitment process pages.

FacePrep - Capgemini pseudo-code section guide, game-based aptitude prep guide, and placement papers pages.

GeeksforGeeks - Capgemini pseudocode quiz bank and candidate interview experience reports.

Game mechanics

JobTestPrep - Aon smartPredict overview, and the individual switchChallenge, gridChallenge and digitChallenge format guides. These describe the Aon, formerly cut-e, engine that supplies the game-based section.

Important caveats

Capgemini does not publish official cut-offs, per-section question counts, or band criteria. Every such figure in this book is an aggregate of candidate reports and should be treated as directional rather than authoritative.

Sources disagree on two points. First, whether English includes a short written component; the essay writing test is reported as removed in the current pattern but is still listed by at least one source. Second, whether Python is permitted in the coding round; the more specific Exceller guidance says C, C++ and Java only. Prepare for the stricter reading of both.

The number of rounds, section timings and exact question counts vary by college and by drive. Confirm the format for your specific drive with your placement cell.

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